MoCHII: Physics, Atomic Data, Algorithms, and Validation

Abstract

MoCHII (MOnte Carlo for H II regions) adds a photoionization gas layer to the validated octree-AMR dust radiative-transfer engine of MoCafe v2.00. This memo documents the ionizing frequency band and its zero-variance transport, the complete atomic-data inventory with sources and the analytic fitting functions evaluated at run time, the iteration and equilibrium algorithms (ionization, thermal balance, trace-metal cascade, explicit diffuse field), the n-level emissivity solver, the ported nebular continuum and Storey & Hummer recombination lines, and the quantitative validation gates: rate integrals vs. analytic attenuation (bias +0.004%); the Strömgren sphere vs. an exact 1D reference (+0.56% in ionized volume; AMR $\equiv$ uniform); the thermal H/He nebula (median 0.14% in T_e); the MOCASSIN Lexington benchmarks (T_e = 8169 K at HII40 and 6925 K at HII20 with the density-suppressed cooling default, matching a current CLOUDY c25.00 run to 0.50% and 0.22%; the zero-density limit gives 7894 and 6376 K, and both modern codes sit above the 2000-era published ranges); and the line-flux comparison (Hβ within a few percent of the published values, the optical collisional lines within the published spread). Also documented: the solution-driven I-front re-refinement (re-refined vs. native high-resolution octree +0.008% in ionized volume with 4.5$\times$ fewer cells) and an Illustris-TNG post-processing demonstration; the completed dust coupling (ionizing-band dust absorption with the dusty Strömgren reduction 0.724 reproduced to +0.34%; EUV/FUV dust *scattering* with forced-first interaction sampling, validated by bracketing; live ionization-dependent dust with a PAH-specific survival factor; and the equilibrium dust temperature + IR spectrum consuming the transported grain heating, energy closure 1.000); and the incremental metal additions (argon, magnesium, iron, and silicon/chlorine/calcium) as pure data operations.

Contents

1. Scope and design rules

MoCHII copies validated modules from MoCafe v2.00 (octree, event-driven traversal, MPI shared memory, HDF5/FITS I/O) and adds gas microphysics under three design rules: (1) *atomic data are fitted offline, evaluated online* — the Python pipeline under `tools/fitting/` is the auditable source of every coefficient in `data/atomic/`, and Fortran only evaluates closed forms; (2) *adding an ion is a data operation*, realized by the species registry (Section 4.6); (3) *validation by quantitative gates*, each capability compared against an analytic or external reference.

The capabilities documented here: the ionizing band + rate integrals; ionization equilibrium + opacity feedback (case B, fixed T_e); thermal balance + trace-metal frame; the explicit diffuse field (case A); solution-driven I-front re-refinement + snapshot post-processing; the dust coupling (absorption competition, scattering, live PAH split, grain heating/temperature, and the SEDust stochastic-emission coupling); the incremental ions (argon, magnesium, iron, silicon, chlorine, calcium); plus the output-time diagnostic layer: n-level emissivities, Storey & Hummer recombination lines, and the nebular continuum.

2. The ionizing band and transport

2.1 Continuous energy and diagnostic bins

The ionizing band covers photon energies $[E_{\rm min}, E_{\rm max}] = [13.598, 100]$ eV. By default (`par%ion_energy_mode = 'continuous'`), a packet energy is sampled from the normalized source-energy CDF, rather than from a frequency bin. For a Planck source the CDF is built adaptively from the Planck energy spectrum; for a tabulated source it is the cumulative integral of the input L_E. H I, He I, He II, and active metal thresholds are retained as knots in the adaptive Planck CDF. With $u \in [0, 1)$, the draw is

$$F_E(E) = \frac{\int_{E_{\rm min}}^{E} L_{E'}\,{\rm d}E'}{\int_{E_{\rm min}}^{E_{\rm max}} L_{E'}\,{\rm d}E'}, \qquad E_{\rm pkt} = F_E^{-1}(u), \qquad L_{\rm pkt} = \frac{L_{\rm band}}{N_{\rm pkt}}. \tag{1}$$

Here $L_{\rm band}$ is the luminosity integrated over the whole ionizing band, not the luminosity of one energy interval. Each packet is therefore an equal-luminosity sample of the source spectrum.

The packet retains $E_{\rm pkt}$ throughout transport. Gas opacity, photoionization cross sections, and the excess energy used for heating are evaluated at this exact energy. The n_ν bins (`par%nnu_ion`; gates use 32) receive only a diagnostic J_E tally: the sampled energy is located by a binary search of the bin-edge array and assigned an output index. Consequently the number or placement of those bins cannot alter the continuous-mode physical solution; it changes only the spectral resolution of saved tallies and rate diagnostics. This is important at thresholds: a photon exactly on a stored edge is assigned to the bin above it, but its interaction physics still uses its own energy.

The validated continuous-energy path currently accepts one internal point source with a Planck or tabulated spectrum. It rejects unsupported combinations (external sources, FUV extension, ionizing dust, secondary ionization, metastable-He, and peel-off options) rather than falling back silently to grouped energy physics.

Grouped mode. With `par%ion_energy_mode = 'grouped'`, packets instead sample a luminosity bin and use its representative energy; this is an explicit compatibility/reproduction mode, not the default physical-energy treatment. In this explicit compatibility mode, `par%ion_align_edges` (default on) pins the ionizing-band bin edges onto the ionization thresholds — H I (13.6 eV), He I (24.6 eV), He II (54.4 eV), plus the active metal photoionization thresholds gathered from the registry — so that no bin straddles a threshold. The sub-segments between consecutive thresholds are log-uniform, and bins are distributed among the segments by log width (largest-remainder, floor 1). When the thresholds outnumber the bins the lowest-priority (metal) edges are dropped gracefully — the H/He edges are always kept — so the option is safe at any `nnu_ion`. Setting it `.false.` uses a single log grid. The alignment removes a real discretization error: a coarse 32-bin log grid straddles the 24.6 eV He I edge and delivers $\sim$18% too many He-ionizing photons against a true 40 kK blackbody ($Q_{\rm HeI}/Q_{\rm HI} = 0.127$ versus the exact 0.108), over-ionizing He (and, downstream, the metals); aligning the edges restores the exact ratio at fixed bin count (Section 5.4).

FUV extension. With `par%add_fuv` the band grows a second, lower segment: `par%nnu_fuv` log bins on $[E_{\rm FUV}, E_{\rm min}]$ (`par%efuv_min`, default 6 eV) *below* the unchanged ionizing bins, so widening the band does not coarsen the ionizing resolution (the earlier single-segment caveat). `par%luminosity` keeps its meaning as the ionizing $[E_{\rm min}, E_{\rm max}]$ luminosity; the FUV part of the same source spectrum rides on top ($L_{\rm FUV}/L_{\rm ion} = 1.097$ for a 4×10^4 K Planck source at 6 eV), so the ionizing photon budget $Q_{\rm H}$ is preserved without manual band-ratio rescaling and packets carry $L_{\rm pkt} = (L_{\rm ion} + L_{\rm FUV})/N$. H and He cross sections vanish below their thresholds, so the FUV bins interact only with dust and with the low-threshold metal stages (next subsection) — they penetrate the neutral gas beyond the I-front and photoionize Mg I (7.65 eV), C I (11.26 eV), S I (10.36 eV), and Fe I (7.90 eV) there, producing the PDR-side Mg II and C II that the ionizing-only band cannot.

2.2 Source components: multiple point sources and an external field

The band is fed by any number of *source components*: `par%nsource` internal point sources (0–16; positions `par%src_x/y/z`, ionizing luminosities `par%src_lum` — all set, or none for an equal split of `par%luminosity`, which becomes the sum) plus, independently, one isotropic external radiation field, on when `par%ext_intensity` > 0. Each packet draws a component in proportion to its band-total luminosity, is emitted from it, and draws its frequency bin from that component's own spectrum CDF; every packet carries $L_{\rm pkt} = \sum_c L_c/N$. A single-component run takes the legacy code path (the RNG stream is unchanged, so earlier results reproduce). The spectrum of internal source i resolves as `par%src_spectrum_file` column i $>$ `src_tstar`(i) Planck $>$ the global `ion_spectrum` file $>$ the global `tstar`; `src_spectrum_file` is one multi-column text file — column 1 the photon energy E [eV] ascending, then one L_E column for each source in arbitrary units, each independently normalized so source i's ionizing segment carries `src_lum`(i).

The external field is an isotropic band-integrated mean intensity J [erg s^{-1} cm^{-2} sr^{-1}] entering through an illuminating surface (`par%ext_geometry`): `'rec'`, the box faces (face-area-weighted — correct for a non-cubic box), or `'sph'`, the bounding sphere of radius `par%rmax` about the box center. The inward flux through the surface is the hemisphere integral of $J\cos\theta$, so the entering band power is

$$F_{\rm in} = \int_{2\pi} J\cos\theta\, d\Omega = \pi J, \qquad L = \pi J A_{\rm surf}. \tag{2}$$

Entry directions are cosine-weighted (Lambert) into the inward hemisphere, $\mu = \sqrt{\xi}$, which reproduces an isotropic interior field with energy density $u = 4\pi J/c$: an optically thin box tallies an interior mean intensity equal to J (the gate of Section 5.8). Its spectrum resolves as `ext_spectrum` (file) $>$ `ext_tstar` Planck $>$ the global spectrum.

Physical-unit spectra. Every file slot is read in the column units set by `par%spectrum_type` (default `'shape'`, arbitrary units renormalized to the scale — the legacy behavior). The physical types `'per_ev'`, `'per_hz'`, `'per_ang'`, `'per_um'` give the first column as E [eV], ν [Hz], or λ [Å or μm] and the value columns as a luminosity density L_X (internal sources) or an intensity density J_X (external field) in the matching spectral unit; wavelength tables are converted to energy and sorted. The `'shape'` value columns are read as the per-eV density L_E on the E [eV] grid; a shape tabulated in wavelength or frequency must use the matching `per_*` type, whose conversion applies the $\mathrm{d}\lambda/\mathrm{d}E$ (or $\mathrm{d}\nu/\mathrm{d}E$) Jacobian, so a raw re-grid onto E under `'shape'` would be wrong by E^{-2}. A physical spectrum is *absolute*: its bin luminosities follow the tabulated integral directly. If the scale (`luminosity` / `src_lum(`i`)` / `ext_intensity`) is unset it is derived from the file and reported; if set, the spectrum is rescaled to it and acts as a shape. With `add_fuv` the same table fills the FUV bins with no separate normalization; a spectrum with no luminosity inside the band aborts.

ISRF presets. `par%ext_spectrum` may instead name one of three analytic interstellar radiation fields: `'draine'` (Draine 1978), `'habing'` (the Draine shape at Habing 1968 normalization, $/1.71$), and `'mathis'` (Mathis, Mezger & Panagia 1983), the shape of the SEDust radiation-field module. They are FUV-only (5–13.6 eV, exactly zero in the ionizing bins), so they require `add_fuv`, and `par%ext_scale` multiplies them. The Draine field is the energy-weighted photon polynomial $F_{\rm ph}(E) = 1.658\times10^6 E^2 - 2.152\times10^5 E^3 + 6.919\times10^3 E^4$ with $J_E = e\,F_{\rm ph}$ ($e = 1.602\times10^{-12}$ erg eV^{-1}; $F_{\rm ph} = E\,N(E)$ is energy-weighted — a common pitfall). The presets carry the canonical FUV energy densities $u_{\rm FUV} = 8.94$, 5.23, 6.01×10^{-14} erg cm^{-3} (Draine / Habing / Mathis; $G_0 = 1.69$, 0.99, 1.14 in Habing units). When `ext_intensity` is set the preset or physical file is rescaled to that band-integrated J (a note is printed; `ext_scale` is then ignored).

2.3 Opacity and the zero-variance edge walk

Each leaf carries the band absorption coefficient per code length

$$\kappa_b \;=\; n_{\rm H}\,[x_{\rm HI}\,\sigma_{\rm HI}(E_b) + y_{\rm He}\,(x_{\rm HeI}\sigma_{\rm HeI} + x_{\rm HeII}\sigma_{\rm HeII})(E_b)]\,d_{\rm cm}\;(+\,\kappa_{\rm dust}), \tag{3}$$

stored as the full $(n_\nu, n_{\rm leaf})$ array `kap_ion` — the grey $s_{\rm ext}$ rescaling of the dust SED band cannot apply because the gas opacity varies with both leaf and bin. With `par%ion_metal_abs` (default on when `use_metals`) the registry metals join the absorption, $\Delta\kappa_b = n_{\rm H}\sum_{\rm el} A_{\rm el}\sum_i f_i\,\sigma_{{\rm VFKY96},i}(E_b)\,d_{\rm cm}$, with the stage fractions f_i from the same product chain as the cooling, iterated by the opacity feedback exactly like $x_{\rm HI}$. Above 13.6 eV this is a $\sim10^{-22}$ cm^2 per H perturbation next to H/He and dust; in the FUV bins it is the only *gas* opacity. The absorbed share returns to the metals through their own Γ integrals, so the bookkeeping stays consistent (metal photoheating is not added to the thermal balance — negligible at trace abundances).

Where the band has no scattering and the diffuse field is emitted explicitly, every packet's contribution to the mean-intensity tally is computed *analytically* along its ray (the forced-first-flight estimator of MoCafe, made exact):

$$\Delta J_b(\text{leaf}) \;=\; w\,L_{\text{pkt}}\,\frac{e^{-\tau_{\text{in}}}-e^{-\tau_{\text{out}}}}{\kappa_b}, \tag{4}$$

accumulated leaf by leaf until the domain edge (`raytrace_ion_to_edge_amr`). No interaction sampling is needed; the only Monte Carlo variance is in direction and bin sampling. The rate-integral gate verifies this estimator to a mean bias of +0.004%.

2.4 Rate integrals

With V the leaf volume and J_b the reduced tally, the code forms for each absorber i

$$\Gamma_i = \sum_b \frac{[4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)}{h\nu_b}, \qquad \mathcal{H}_i = \sum_b [4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)\left(1-\frac{E_{\text{th},i}}{E_b}\right), \tag{5}$$

the photoionization rate [s^{-1}] and photoheating [erg s^{-1}] for each particle of species i.

3. Atomic data

Every coefficient file in `data/atomic/` carries a provenance header (source, version, fit range, maximum fit error). The generators live in `tools/fitting/`.

3.1 Photoionization cross sections

The one-electron ions H I (Z=1) and He II (Z=2) use the *exact* ground-state hydrogenic cross section (Osterbrock & Ferland 2006, Eq. 2.4; `sigma_hydrogenic`, the same form as EXHALE):

$$\varepsilon = \sqrt{E/E_{\text{th}}-1}, \qquad \sigma(E) = \frac{\sigma_0}{Z^2}\left(\frac{E_{\text{th}}}{E}\right)^4 \frac{\exp\!\left(4-4\arctan\varepsilon/\varepsilon\right)}{1-e^{-2\pi/\varepsilon}}, \tag{6}$$

with $\sigma_0 = 6.30\times10^{-18}\,\text{cm}^2$ the threshold value and E_{th} the true ionization potential (13.598 and 54.416 eV), tied to the physical edge (not $13.6Z^2$) so it stays aligned with the band grid. He I (two electrons) and every metal ion use the full VFKY96 single-shell analytic form (Verner, Ferland, Korista & Yakovlev 1996, ApJ 465, 487, Eq. 1):

$$x = \frac{E}{E_0}-y_0, \quad z = \sqrt{x^2+y_1^2}, \quad Q = 5.5-\tfrac{1}{2}P, \quad \sigma(E) = \sigma_0\left[(x-1)^2+y_w^2\right] z^{-Q}\left(1+\sqrt{z/y_a}\right)^{-P}, \tag{7}$$

zero below threshold, σ_0 in Mb. Parameters were parsed directly from the authors' `phfit2.f` `PH2` (outer-shell) table and verified against the EXHALE implementation; H/He values are compiled in `photo_xsec.f90`, the registry ions in `element_<el>.txt`. For the elements *absent* from the VFKY96 outer-shell table (P, Cl, K — Opacity-Project gaps; the MOCASSIN `ph2.dat` read loop skips the same Z), the Verner & Yakovlev (1995) outer-shell subshell fit is used instead (`sigma_vy95`; `PHOTO2` lines): $\sigma = \sigma_0[(y-1)^2+y_w^2]\,y^{-(5.5+l-P/2)}(1+\sqrt{y/y_a})^{-P}$ with $y = E/E_0$ and l the subshell orbital quantum number — the same VFKY96/VY95 hybrid as MOCASSIN `phFitEl`. Chlorine is the first user of this path. Among H/He, only the He I ground state uses VFKY96:

ion	$E_{\rm th}$[eV]	E_0	σ_0[Mb]	y_a	P	y_w	y_0	y_1
He I	24.587	13.61	949.2	1.469	3.188	2.039	0.4434	2.136

plus C I–III, N I–II, O I–III, Ne I–III, S I–IV, Ar I–IV (20 rows, in the element files). The rate-integral gate cross-checks the curves against EXHALE's independent implementations. The exact H I and He II cross sections give the threshold values $\sigma_0/Z^2 = 6.30$ and 1.575 Mb; the VFKY96 fits they replace for H/He agreed with the exact curve to 0.73% (H I) and 0.43% (He II), and $\sigma_{\rm HeI} = 7.43$ Mb (VFKY96).

3.2 H/He recombination

Recombination coefficients in the ionization balance are chosen by `par%recomb_model`. The default `'badnell_milne'` takes the total (case A) radiative recombination α_A from the same Badnell (2023) RR fit used for the metals [Eq. (12)], obtains the ground-level (n=1) direct-capture rate α_1 from the Milne relation applied to the *same* photoionization cross sections the transport absorbs with [Eq. (28)], and forms case B by removing the ground term, $\alpha_B = \alpha_A - \alpha_1$. At run time α_1 is evaluated from the seven-coefficient functional form of Mao & Kaastra (2016, A&A 587, A84) ($T_{\rm eV} = k_B T$; a_0 in $10^{-10}\,{\rm cm^3\,s^{-1}}$),

$$\alpha_1(T) = a_0\,10^{-10}\,T_{\rm eV}^{-b_0-c_0\ln T_{\rm eV}}\,\frac{1+a_2\,T_{\rm eV}^{-b_2}}{1+a_1\,T_{\rm eV}^{-b_1}}, \tag{8}$$

but the coefficients are fitted to that Milne integral instead of being taken from Mao & Kaastra; §4.8 lists them with their fit residuals and gives the measurement that forced the replacement. The routine name `rr_mao` in `recomb_mod` therefore records the functional form, not the origin of the numbers, while the namelist value names the two determinations the default combines — Badnell's total and the Milne ground level — and so records that α_B here is a difference rather than a fit of its own. Because α_1 and the ground-continuum emissivity of §4.8 are now one integral over one cross section, the case A/B split and the diffuse packet count cannot drift apart. He II→He I adds the Badnell three-term dielectronic sum to α_A (negligible below $\sim 5\times10^4$ K); H II and He III are bare nuclei with no DR. The coefficients at 10^4 K ($10^{-13}\,{\rm cm^3\,s^{-1}}$), with the independently level-resolved Cloudy c25.00 values in parentheses, are

reaction	α_A	α_1	α_B
H II→H I	4.19 (4.16)	1.579 (1.584)	2.61
He II→He I	4.37 (4.36)	1.609 (1.603)	2.76
He III→He II	21.9 (21.8)	6.512 (6.515)	15.4

Setting `par%recomb_model = 'hui_gnedin'` restores the earlier Hui & Gnedin (1997, MNRAS 292, 27) case-A/B fits with $\lambda = 2T_{\rm TR}/T$ ($T_{\rm TR}$ = 157807, 285335, 631515 K for H I, He I, He II):

$$\alpha_A^{\rm HII} = 1.269\times10^{-13}\,\lambda^{1.503}\big/\left[1+(\lambda/0.522)^{0.470}\right]^{1.923}, \qquad \alpha_B^{\rm HII} = 2.753\times10^{-14}\,\lambda^{1.500}\big/\left[1+(\lambda/2.740)^{0.407}\right]^{2.242},$$
$$\alpha_A^{\rm HeII} = 3.000\times10^{-14}\,\lambda^{0.654}, \qquad \alpha_B^{\rm HeII} = 1.260\times10^{-14}\,\lambda^{0.750}, \tag{9}$$

He III hydrogenic (Z=2). These are retained so the recorded gates reproduce; they agree with the default to $<0.6\%$ for H II and He III, but give He II→He I 5% lower ($\alpha_B = 2.62$ vs $2.76\times10^{-13}\,{\rm cm^3\,s^{-1}}$). Recombination *cooling* uses the companion fits

$$\beta_B^{\rm HII} = 3.435\times10^{-30}\,T\,\lambda^{1.970}\big/\left[1+(\lambda/2.250)^{0.376}\right]^{3.720}\ {\rm erg\,cm^3\,s^{-1}}, \tag{10}$$

(case A analog with 1.778×10^{-29}, 0.541, 1.965, 0.502, 2.697), He III scaled as for α, and He II as $k_B T\,\alpha_{A,B}^{\rm HeII}$ (their recommendation; He II dielectronic recombination is negligible at nebular temperatures).

3.3 H/He microphysics: comparison with other codes

The H/He cross sections and recombination can be placed next to three reference codes. For the photoionization cross section, MoCHII and EXHALE evaluate the exact hydrogenic formula [Eq. (6)] for H I/He II; MOCASSIN fits them with VFKY96 (agreeing with the exact curve to $< 1\%$), and Cloudy resolves them from its iso-electronic H-like/He-like model atoms. For the recombination that closes the ionization balance (Table 1), MoCHII's default takes α_A from the modern Strathclyde (Badnell) data and splits off α_1 with the Milne relation, which is how Cloudy resolves its H-like and He-like recombination level by level; MOCASSIN uses the older Verner & Ferland (1996) total; EXHALE, a planetary-atmosphere code, adopted the Badnell RR + Mao & Kaastra α_1 construction as its default on 2026-07-17, before the α_1 correction of §4.8 (Hui & Gnedin remains a legacy option). MOCASSIN and Cloudy always work in case A and transport the diffuse ground-recombination field explicitly, whereas MoCHII defaults to case B (on-the-spot) with an optional explicit diffuse field, and EXHALE is case B only.

Table 1: H/He recombination in the ionization balance: source and α at 10^4 K ($10^{-13}\,\mathrm{cm^3\,s^{-1}}$; case A / case B where both apply).

code	source	H II→I	He II→I	He III→II
MoCHII (default)	Badnell RR + Milne α_1	4.19 / 2.61	4.37 / 2.76	21.9 / 15.4
MoCHII (`hui_gnedin`)	Hui & Gnedin 1997	4.30 / 2.59	4.23 / 2.62	22.3 / 15.5
EXHALE (default)	Badnell RR + Mao & Kaastra α_1	– / 2.61	– / 2.81	– / 15.4
EXHALE (`legacy_hhe_rates`)	Hui & Gnedin 1997 (case B)	– / 2.59	– / 2.62	– / 15.5
MOCASSIN	Verner & Ferland 1996 (case A)	4.19	4.84	21.9
Cloudy c25.00	level-resolved Milne, own $\sigma_{\rm pi}$ (case A)	4.16	4.36	21.8

The H II and He III rates agree across all four codes to within a few percent. The one substantive difference is He II→He I: MoCHII's Badnell total (4.37) and Cloudy's explicit sum over levels (4.36) agree to 0.3%, MOCASSIN's Verner & Ferland value (4.84) is the high outlier, and the earlier Hui & Gnedin (4.23) was low. Moving MoCHII onto the modern Badnell value therefore raises its case-B He recombination by $\sim 5\%$, in the direction that reduces the He over-ionization seen relative to converged MOCASSIN in the HII40 benchmark (§5. discusses that residual). Because that 11% spread in a single coefficient is the largest code-to-code difference anywhere in the H/He data, §3.4 takes it apart.

3.4 The helium ionization network, code by code

Neutral helium is the one species in the H/He inventory on which the photoionization codes disagree at more than the percent level. Reading each code's own data rather than its published description gives, for α_A(He I) at 10^4 K,

code	how α_A(He I) is obtained	$10^{-13}\,\mathrm{cm^3\,s^{-1}}$	vs. Cloudy
Cloudy c25.00	level-by-level Milne from its own σ_{pi}, total cached	4.3583	—
MoCHII	Badnell RR total + DR	4.3719	+0.3%
Wood `ionize` / CMacIonize	sum of the four BSS99 *effective* coefficients	4.2630	−2.2%
MOCASSIN	Verner & Ferland 1996 total fit	4.8394	+11.0%

The MOCASSIN entry is decoded from `data/radrec.dat`, replicating the read order of `update_mod.f90` lines 1707–1735; the decode is validated on the two channels whose answer is known independently, giving H I 4.1923 against Cloudy's 4.1642×10^{-13} (+0.7%) and He II 2.1880 against 2.1817×10^{-12} (+0.3%).

The spread is not the extent of the cascade sum. A natural first reading of an 11% spread is that the codes stop the recombination cascade at different levels. None of them does. Three treatments are in play and all three are complete: Cloudy sums explicitly over resolved levels; MoCHII and MOCASSIN use totals over all n (Badnell, VF96); Wood and CMacIonize use *effective* coefficients, in which the cascades from higher levels are already folded into the $n = 2$ terms. MoCHII's Badnell total agrees with Cloudy's explicit level sum to 0.3%, which is what a truncation on either side would break. (The 20-level list in Cloudy's `source/sanity_check.cpp` is a unit test of threshold cross sections, not the extent of its recombination sum.)

The control that settles it is the electron count:

species	electrons	code-to-code spread in α_A
H I	1 (hydrogenic)	0.7%
He II	1 (hydrogenic)	0.3%
He I	**2**	**11–13%**

The one-electron species have an exact analytic solution, so there is nothing for two calculations to disagree about, and they do not; the two-electron one has no closed form and carries screening and the singlet/triplet structure, and it spreads by 11–13%. The disagreement is therefore in the atomic data for a two-electron atom, not in how far anybody sums.

Three of the four cluster. Within He I the values are not scattered but split three-to-one: Cloudy's level sum 4.3583, Badnell 4.3719 and BSS99 $4.2630\times10^{-13}\,\mathrm{cm^3\,s^{-1}}$ all lie within 2.5% of each other, and Verner & Ferland (1996) alone sits at 4.8394×10^{-13}. MOCASSIN applies its `radRecFit` uniformly with no special case for H or He (`source/update_mod.f90:1636--1639`, whose own comment names "H-like, He-like, Li-like, Na-like — Verner & Ferland, 1996, ApJS, 103, 467"), and adds dielectronic recombination only for $Z_{\mathrm{elem}} \geq 3$, so its He I rate is the VF96 radiative value alone. Offered as reasoning rather than as a verified claim: VF96's He-like fits lean on hydrogenic scaling for the excited levels, and $Z = 2$ is where that approximation is worst, since neutral helium is exactly where screening and the singlet/triplet splitting matter most.

A caution for the MOCASSIN regression suite. MOCASSIN's 11-case suite is used here as a three-dimensional validation reference. Every helium comparison against it inherits a known +11% in α_A(He I), and that difference has been traced to VF96 by the first-principles Cloudy determination above, so it should be subtracted before a disagreement in a helium ionization front or a helium line intensity is attributed to transport or geometry. The same comparison across the metal registry, where no such independent determination is available, is in §3.7.

Excited He I levels are not a hidden channel. What Cloudy's resolved He I model atom has and an effective treatment does not is a population in the $n = 2$ terms that can itself be ionized. MoCHII cannot photoionize those levels at all: the band floor is `par%eion_min` = 13.598 eV while every excited He I level has an ionization potential below it (2^3S 4.767, 2^1S 3.971, 2^3P 3.623, 2^1P 3.369 eV). The size of what is missed is bounded by the metastable population, since 2^1S decays by two-photon emission at 51.3 s^{-1} and is left 10^7 times less populated than 2^3S: with $n(2^3\mathrm{S})/n(\mathrm{He}^0) = 6.2\times10^{-5}$ at $r = 3$ pc, $Q(>4.767\ \mathrm{eV})/Q(>24.587\ \mathrm{eV}) = 33.8$ for a 40-kK blackbody and Cloudy's own 2^3S threshold cross section of 5.48×10^{-18} cm^2, the extra ionization is 0.2–0.6%, and collisional ionization out of 2^3S adds a comparable ∼0.6%. Taken together with the front sensitivity measured in §4.8 (0.80 pc per unit fractional change in the He I recombination rate), the 11–13% code spread is worth 0.09 pc — half the HII40 front residual against Cloudy — but MoCHII itself lies within 0.3% of Cloudy, so that lever is not available to MoCHII: it measures how much of a helium front comparison between two arbitrary codes can be recombination data alone.

3.5 Collisional ionization (Voronov 1997)

$$k(T) = A\,\frac{1+P\sqrt{U}}{X+U}\,U^K e^{-U}, \qquad U = \frac{\Delta E}{kT}, \tag{11}$$

with parameters (ΔE [eV], P, A, X, K) parsed from the author's `cfit.f` table:

ion	ΔE	P	$A\,[\mathrm{cm^3\,s^{-1}}]$	X	K
H I	13.6	0	2.91×10^{-8}	0.232	0.39
He I	24.6	0	1.75×10^{-8}	0.180	0.35
He II	54.4	1	2.05×10^{-9}	0.265	0.25

plus the 16 registry rows in the element files. An optional `par%ci_model = 'dere_hybrid'` multiplies each Voronov rate by the constant Dere (2007)/Voronov ratio per (element, stage) taken from Cloudy c23.01's `DereRatio` table; the default `'voronov'` leaves the rate unchanged. In photoionized gas at $\sim10^4$ K the metal collisional ionization is Boltzmann-suppressed far below photoionization, so the hybrid shifts the ionization structure only in shielded, low-ionization zones where collisions are the sole metal-ionization channel.

3.6 Metal recombination: Badnell RR + DR

For the registry cascade, total recombination of the recombining ion is Badnell (2006, ApJS 167, 334) radiative recombination

$$\alpha_{\mathrm{RR}}(T) = A\left[\sqrt{T/T_0}\,\left(1+\sqrt{T/T_0}\right)^{1-b}\left(1+\sqrt{T/T_1}\right)^{1+b}\right]^{-1}, \qquad b = B + Ce^{-T_2/T}, \tag{12}$$

plus the Badnell-group dielectronic fits $\alpha_{\mathrm{DR}}(T) = T^{-3/2}\sum_i c_i e^{-E_i/T}$. Coefficients are read from the Badnell/Strathclyde AMDPP TAMOC tables (the `clist_K` files, 2023 release, copied into `data/atomic/badnell_{rr,dr}.dat` and verified md5-identical to the copies Cloudy c23.01 redistributes), using the M=1 ground-level fit of each recombining ion. Most ions are numerically identical to the CHIANTI v11.0.2 `.rrparams/.drparams` (which *are* Badnell data), the notable exception being Si II: CHIANTI 11 still carries the Abdel-Naby et al. (2012) DR, whose $\sim 10^4$-K resonance is stronger, so the 2023 refit lowers the total $\mathrm{Si}^+ + e \to \mathrm{Si}^0$ recombination to $0.48\times$ the CHIANTI value at 10^4 K. Ions Badnell does not tabulate (the low Ar/Fe/Cl/Ca stages, confirmed absent at every M) are taken from the CHIANTI files instead. Where CHIANTI carries no Badnell DR (Ar II, Ar III, Cl II, Ca II, Ca IV, Ca V), the `.drparams` type-2 entries are the Shull & Van Steenberg (1982) form $\alpha_{\rm DR} = A\,T^{-3/2} e^{-T_0/T}(1 + B\,e^{-T_1/T})$, written as `DR2` lines and supported by the same reader (negligible at nebular temperatures; kept for completeness). One further CHIANTI RR type appears with iron and the higher calcium stages: the Fe II–IV, Ca IV and Ca V `.rrparams` are type-3 power laws, written as `RR2` lines and evaluated as $\alpha_{\rm RR} = A\,(T/10^4\,{\rm K})^{-\eta}$. Section 3.7 audits all of this entry by entry.

3.7 Recombination data across the registry: an audit

The α_1 source-mixing defect of §4.8 was found in the H/He channels, and it prompted a sweep of every recombination coefficient the code carries — 32 registry entries over 11 elements, plus the hardcoded H/He channels. Three questions were put to them: is each coefficient the one its provenance header claims, is the functional form the best available for that ion, and how far does the answer depend on which data set is used.

Transcription: no errors. Of the 32 registry entries, 22 take their coefficients from the Badnell tables, and all 22 reproduce `data/atomic/badnell_{rr,dr}.dat` at 10^4 K to a ratio of 1.0000. The key is fixed by the registry convention: `TRANSITION` i of element Z is the recombination of the ion of charge i, so the Badnell row is (Z, $N = Z - i$, $M = 1$). MoCHII's hardcoded H I coefficients (8.318×10^{-11}, 0.7472, 2.965, 7.001×10^5) are exactly the $Z = 1$, $N = 0$ row of the same table. There is no transcription error anywhere in the registry, and no coefficient traceable to the wrong ion.

The older forms cannot be migrated, and should not be. Badnell's tabulation covers $N \leq 15$ (through the Al-like sequence) plus $N = 18$ (Ar-like) for $Z \geq 20$. Every use of `RR2` or `DR2` in the registry falls at a (Z, N) outside that coverage. Where Badnell has nothing, `tools/fitting/make_element_data.py` takes CHIANTI v11.0.2 and writes *exactly the functional form CHIANTI declares*: `rrparams` types 1 and 2 are the Badnell form and become `RR`, type 3 is the power law and becomes `RR2`; `drparams` type 1 becomes `DR`, and type 2, the Shull & Van Steenberg (1982) form, becomes `DR2`. Checked against the CHIANTI file of the recombining ion, entry by entry:

registry entry	recombining ion	N	Badnell	CHIANTI `rr`/`dr` type → form
Ar 1	Ar II	17	absent	2 / 2 → `RR` + `DR2`
Ar 2	Ar III	16	absent	1 / 2 → `RR` + `DR2`
Ar 3	Ar IV	15	covered	2 / 1 → `RR` + `DR`
Cl 1	Cl II	16	absent	1 / 2 → `RR` + `DR2`
Ca 1	Ca II	19	absent	1 / 2 → `RR` + `DR2`
Ca 2	Ca III	18	covered	2 / 1 → `RR` + `DR`
Ca 3	Ca IV	17	absent	3 / 2 → `RR2` + `DR2`
Ca 4	Ca V	16	absent	3 / 2 → `RR2` + `DR2`
Fe 1, 2, 3	Fe II–IV	25, 24, 23	absent	3 / 1 → `RR2` + `DR`

Ar IV and Ca III are the controls: Badnell covers $N = 15$, and $N = 18$ for $Z \geq 20$, and the registry duly uses `RR` + `DR` at both. So the Shull & Van Steenberg form is not a stale choice made in preference to Badnell — it is what CHIANTI itself still ships for those ions. There is nothing to migrate, and the registry already uses the best source available at every element and stage. A suggestion recorded earlier, to move the `RR2` and `DR2` entries onto Badnell, is therefore withdrawn on both counts: Badnell has no row for them, and CHIANTI's current data for them is in those forms.

Code to code: variance between data sets, not an error bar on MoCHII. What transcription cannot settle is which data set is right. Comparing radiative recombination only — dielectronic recombination is excluded on both sides, because MOCASSIN adds it separately and keeping it on one side alone would compare unlike quantities — MoCHII's Badnell/CHIANTI values against the Verner & Ferland (1996) fits MOCASSIN uses, at 10^4 K:

recombination forming	MoCHII / VF96
C II, C III, N II, Ne II	0.999–1.016
N I	0.971
Si IV (Na-like)	0.932
Ne I	0.898
C I	0.861
O I	0.837
Mg II (Na-like)	0.791

Eleven ions: seven within 10%, ten within 20%, median 0.971. The spread is concentrated in the rates that form *neutral atoms*, which is where He I sits as well, at 11% (§3.4).

What that table is has to be stated precisely. It is the variance between two data sets, *not* a measured error in MoCHII, and for the metals there is no third arbiter available here: Cloudy ships `data/badnell_rr.dat` and `badnell_dr.dat` md5-identical to MoCHII's copies and evaluates them with `RR_Badnell_rate_coef` and `DR_Badnell_rate_coef` in `source/ion_recomb.cpp`, so Cloudy and MoCHII agree on metal recombination by construction and Cloudy cannot adjudicate. With only two determinations in play, which one is right for C I, O I, Ne I or Mg II is undetermined.

He I is the one exception, and there MoCHII is demonstrably right. Cloudy does *not* use Badnell for the He-like and H-like sequences; it derives them level by level from its own photoionization cross sections through the Milne relation, which is an independent first-principles determination. Against it, Badnell (MoCHII) is +0.3% and VF96 (MOCASSIN) is +11.0% (§3.4), so for He I the spread is attributable to VF96. One may reason by analogy that the metal neutrals behave the same way — Badnell (2006) is the later and more careful calculation, VF96 (1996) leans on simplified treatments of many-electron systems, and He I is the case where that was actually demonstrated — but that is inference, not measurement. Settling it needs an independent determination for each ion: R-matrix values for C I, O I and Ne I recombination, or CHIANTI totals that do not derive from Badnell.

An earlier phrasing, "a 10–20% uncertainty in MoCHII's neutral-forming recombination rates", is withdrawn as imprecise: it reads a spread between two data sets as an error bar on one of them. The two accurate statements are that MoCHII's He I rate agrees with an independent first-principles calculation to 0.3% while the VF96 set MOCASSIN uses is 11% high (measured), and that for the metal neutrals Badnell and VF96 differ by 10–16% with no third source in this configuration able to say which is right (a code-to-code spread).

What the audit establishes, and what it does not. It found no defect in MoCHII's recombination data: no transcription error, no form that a better source could replace, and no coefficient attached to the wrong ion. That is not the same statement as "the data are accurate". For H and He an independent determination exists and MoCHII matches it to 0.3–0.7%; for the metals no arbiter is available in this configuration, so the verdict there is the absence of a defect found, not accuracy established.

3.8 Charge exchange with hydrogen

First-stage reactions use the verified transcription of Huang et al. (2023, ApJ 951, 123) Table 4 (sources Stancil et al. 1998/1999, Lin et al. 2005, Zhao et al. 2005, Kingdon & Ferland 1996), e.g. for oxygen

$$k(\mathrm{O}+\mathrm{H}^+) = 2.08\times10^{-9}\,T_4^{0.405} + 1.11\times10^{-11}\,T_4^{-0.458},$$
$$k(\mathrm{O}^+ + \mathrm{H}) = \left(1.26\times10^{-9}\,T_4^{0.517} + 4.25\times10^{-10}\,T_4^{0.00669}\right) e^{-227/T}. \quad (13)$$

Higher stages ($\mathrm{X}^{i+}+\mathrm{H}^0 \to \mathrm{X}^{(i-1)+}$, recombination direction) use the Kingdon & Ferland (1996) fits $k = a\,T_4^b\,(1+c\,e^{d\,T_4})$, transcribed from the MOCASSIN `update_mod.f90` `chex` table and evaluated at T clamped into the fit validity ($\sim$5 000–50 000 K). *Stage mapping (corrected 2026-07-13).* MOCASSIN's $\mathtt{chex}(Z,\mathrm{ion})$ is the reaction $\mathrm{X}^{\mathrm{ion}+}+\mathrm{H}^0 \to \mathrm{X}^{(\mathrm{ion}-1)+}$, and its inline comment labels the *product* (lower) ion. The recombining ion of transition i is X^{i+}, so the entry is $\mathtt{chex}(Z,i)$. An earlier version read the product-labeled comment as the reactant and used $\mathtt{chex}(Z,i{+}1)$, placing every higher-stage rate one ionization stage too low; the corrected assignment (verified against MOCASSIN's own ratio loop and CMacIonize) is below.

reaction	a [10^{-9}cm^3 s^{-1}]	b	c	d
C^{2+} + H	1.67×10^{-4}	2.79	304.72	−4.07
C^{3+} + H	3.25	0.21	0.19	−3.29
N^{2+} + H	0.305	0.60	2.65	−0.93
O^{2+} + H	1.04	0.27	2.02	−5.92
Ne^{2+} + H	10^{-5}	0	0	0
S^{2+} + H	10^{-5}	0	0	0
S^{3+} + H	2.29	0.0402	1.59	−6.06
Ar^{2+} + H	10^{-5}	0	0	0
Ar^{3+} + H	4.57	0.27	−0.18	−1.57
Si^{2+} + H	1.23	0.24	3.17	4.18×10^{-3}
Si^{3+} + H	0.49	−0.0874	−0.36	−0.79
Si^{4+} + H	7.58	0.37	1.06	−4.09
Mg^{2+} + H	8.58×10^{-5}	2.49×10^{-3}	0.0293	−4.33
Fe^{2+} + H	1.26	0.0772	−0.41	−7.31
Fe^{3+} + H	3.42	0.51	−2.06	−8.99

Mg and Fe were already correct: their Huang Table 4 reactant labels happen to land on $\mathtt{chex}(Z,i)$. Silicon's first stage uses the Huang A9/A10 fits ($\mathrm{Si}+\mathrm{H}^+$ and $\mathrm{Si}^+ + \mathrm{H}^0$, the latter 2.371×10^{-12}cm^3 s^{-1} at 10^4 K); Cl and Ca have no KF96/Huang transcription. The Mg and Fe first-stage reactions follow Huang Table 4: Mg + H^+ is the KF96 form with an added reverse barrier $e^{-6.91/T_4}$ (endothermic); Fe + H^+ is a fast constant 4.0×10^{-9}cm^3 s^{-1} with reverse $a(T/300)^b\, e^{-6.61/T_4}$. The correction raises the HII40 doubly/triply-ionized fractions (Section 5.4); the earlier good agreement of some metal fractions with the underconverged MOCASSIN baseline was partly fortuitous.

With the shipped abundances 29 of these reactions are active — 7 in the $H^+ \to H^0$ direction (C, N, O, Mg, Si, S, Fe, all at the neutral stage) and 22 in the reverse — and every one of them enters hydrogen's balance as well as the metal's (§4.4, `par%charge_exchange`).

3.9 Cooling fits

Collisional line cooling for the thermal loop uses the EXHALE multi-exponential form fitted to direct CHIANTI v11.0.2 evaluations (`fit_cooling.py`; reader `chianti_cooling.py` parses `.elvlc`/`.scups` with Burgess & Tully 1992 descaling):

$$\Lambda_{\rm ion}(T) = T^{-1/2}\sum_{i=1}^{N} A_i\, e^{-T_i/T} \qquad [\mathrm{erg\,cm^3\,s^{-1}\ per\ } (n_e n_{\rm ion})], \tag{14}$$

evaluated in the optically thin $n_e \to 0$ limit of statistical equilibrium: every ion sits in its lowest level, excitation happens out of that level only, and the cascade radiates the full excitation energy, so $\Lambda_{\rm ion}$ is the excitation power out of the ground level and equals what the n-level solve (Section 3.10) returns as $n_e \to 0$. Transition energies are the observed `.elvlc` level differences in both the Boltzmann factor and the radiated photon (the `.scups de` is the Burgess–Tully scaling energy, and detailed balance ties the excitation rate to the level gap that appears in the equilibrium matrix). Fit range 10^3–10^5 K; Levenberg–Marquardt on $\log\Lambda$ with the T_i ladder derived from the excitation temperatures $\Delta E_{1u}/k$ of each ion.

ion	N	max err	ion	N	max err
H I	4	0.52%	O I	6	0.20%
C I	6	0.27%	O II	4	0.27%
C II	5	0.35%	O III	5	0.32%
C III	6	0.50%[a]	S II	3	2.23%
N I	5	0.58%	S III	5	0.23%
N II	5	0.66%	S IV	3	0.61%
N III	5	0.70%	Ne II	5	0.40%
			Ne III	5	0.34%
			Ar III	2	0.22%[b]
			Ar IV	4	0.27%

[a]in the relevant range ($\Lambda > 10^{-3}$ max); 5.5% at the negligible low-T tail. [b]at 10^4 K; 3.9% at the 10^5-K endpoint.

By construction Eq. (14) carries no collisional de-excitation, so it is exact for lines whose critical density is far above the local n_e and increasingly too large for those below it — the low-$n_{\rm crit}$ lines [C II] 158 μm, [O III] 52/88 μm and the [O II]/[S II] optical doublets, whose electron critical densities are 10^2–$10^{3.5}$ cm^{-3}, so at the $n_e \sim 10^2$ cm^{-3} of an H II region they overcool the line budget by $\sim$12% (measured against a CLOUDY c25.00 cooling decomposition on a fixed state). The thermal loop therefore treats Eq. (14) as the $n_e \to 0$ *carrier* and suppresses it to the local density. A

(T, n_e) table, built once at setup (`nlevel_cooling_mod`) by solving the same n-level atom over a grid ($\log_{10} T$ = 2–6, $\log_{10} n_e$ = −6–10), stores for each ion the suppression ratio

$$S_{\rm sup}(T,n_e) = \frac{\Lambda_{\rm SE}(T,n_e)}{\Lambda_{\rm SE}(T,n_e \to 0)} \in (0,1], \qquad \Lambda_{\rm SE}(T,n_e) = \frac{1}{n_e}\sum_k n_u^{(k)} A_k\, \Delta E_k, \tag{15}$$

and the low-density reference $\Lambda_{\rm SE}(T, n_e \to 0)$; the runtime cooling coefficient combines the two as

$$\Lambda(T,n_e) = \Lambda_{\rm fit}(T)\big[f_{\rm cov}(T)\,S_{\rm sup}(T,n_e) + (1 - f_{\rm cov}(T))\big], \qquad f_{\rm cov} = \min\Big(1,\ \tfrac{\Lambda_{\rm SE}(T,n_e\to 0)}{\Lambda_{\rm fit}(T)}\Big). \tag{16}$$

Here $f_{\rm cov}$ is the fraction of the low-density cooling the (often truncated) n-level model resolves; the remainder $1 - f_{\rm cov}$ — the high-excitation resonance lines, of far higher critical density — is left in its low-density limit, so a truncated model cannot bias the total. As $n_e \to 0$, $S_{\rm sup} \to 1$ and $\Lambda \to \Lambda_{\rm fit}$ *exactly*, and an ion with no n-level model would keep the plain fit — there is none left, since C I and N I, the last two gaps, now carry models of their own (their far-infrared fine-structure lines have $n_{\rm crit} \sim 2$–$10\,{\rm cm^{-3}}$, so the suppression is severe: $S_{\rm sup} = 0.43$ at 10^4 K and $n_e = 10^4\,{\rm cm^{-3}}$). This density-suppressed cooling (`par%cooling_model = local_ne`, the default) is built from the very solve that produces the diagnostic lines (Section 3.10), so the thermal balance and the emergent line list are one physics; the pure zero-density fit is retained as the `low_density` option, exact in the diffuse gas the limit is written for. The density-dependent emissivities themselves come from the n-level data.

3.10 n-level (diagnostic) data

For output-time diagnostics, `fit_nlevel.py` exports for each ion the lowest n levels (energies, statistical weights), the radiative A-values (`.wgfa`), and the effective collision strengths $\Upsilon(T)$ as low-order Chebyshev fits in Burgess–Tully scaled space: with $e_t = kT/\Delta E$,

$$s = \begin{cases} 1 - \ln C/\ln(e_t + C) & \text{types 1, 4} \\ e_t/(e_t + C) & \text{types 2, 3, 5, 6} \end{cases}, \qquad u = \frac{2(s - s_{\rm lo})}{s_{\rm hi} - s_{\rm lo}} - 1, \qquad y = \sum_k c_k T_k(u), \tag{17}$$

descaled by type ($y\ln(e_t{+}e)$; y; $y/(e_t{+}1)$; $y\ln(e_t{+}C)$; y/e_t; 10^y), with a log flag storing $\log_{10} y$ for weak peaked intercombination transitions. The n-level export now covers the 24 line-emitting ions of the eleven-element registry (Table 2); the worst transition fit error is 1.7% (O III $^1S_0$–$^5S_2$, $\Upsilon \sim 10^{-4}$), typically $\lesssim 0.5\%$. Ar II has no level/collision data in CHIANTI v11 (only RR/DR), so that stage is tracked without cooling or lines ([Ar II] 6.98 μm omitted). Verification is layered: the fitted files reproduce direct CHIANTI spline evaluation of the classic ratios to 0.05–0.5%; the PyNeb cross-check gives [O III] 5007/4363 within 1.2% and [N II] 6584/5755 within 0.3% (the density diagnostics differ by 6–10% at $10^4\,{\rm cm^{-3}}$ from PyNeb's independent collision data — database spread); and the Fortran solver (Section 4.16) matches the Python reference implementation to four significant digits on 26 lines.

Table 2 lists the principal collisional diagnostic lines each metal block produces (the complete transition list for each ion is written to `<base>_lines.txt`); only active-registry ions with an `nlevel_<el>_<stage>` file get a block. The H I/He II/He I recombination-line set is documented separately in the recombination-lines subsection below.

Table 2: Metal collisional diagnostic lines from the n-level solve (block emis_<el>_<stage> in <base>_emis; principal lines, Å unless μm noted). C, N, O, Ne, S are active by default; Ar, Mg, Fe, Si, Cl, Ca are activated by par%abund_<El>.

ion (block)	principal lines
c_2 [C II]	2325 (UV multiplet), 158 μm
c_3 C III]	977, 1907, 1909, 178 μm
n_2 [N II]	3064/3071, 5756, 6550, 6585, 122 μm, 205 μm
n_3 [N III]	1750 (UV multiplet), 57 μm
o_1 [O I]	5579, 6302, 6366, 63 μm, 145 μm
o_2 [O II]	2471, 3727, 3730, 7321/7331
o_3 [O III]	1661/1666, 2322/2332, 4364, 4960, 5008, 52 μm, 88 μm
ne_2 [Ne II]	12.8 μm
ne_3 [Ne III]	3343, 3870, 3969, 15.55 μm, 36 μm
s_2 [S II]	4070/4078, 6718, 6733, 1.03 μm, 315 μm
s_3 [S III]	3722, 6314, 8831, 9070, 9532, 12 μm, 18.7 μm, 33.5 μm
Optional — activate with par%abund_<El>	
ar_3 [Ar III]	3006/3110, 5193, 7138, 7753, 8.99 μm, 21.8 μm
ar_4 [Ar IV]	2854/2869, 4713, 4741, 7170 (multiplet), 56 μm, 78 μm
mg_2 Mg II	2796, 2804
fe_2 [Fe II]	1.26/1.32 μm, 5.34 μm, 17.9 μm, 26 μm (40 lines total)
fe_3 [Fe III]	4659, 4703, 4881, 5011, 22.9 μm (38 lines; 16/34-level with fe_levels_full)
si_2 [Si II]	2335 (multiplet), 34.8 μm
si_3 Si III]	1206, 1883, 1892, 38 μm
si_4 Si IV	1394, 1403
cl_2 [Cl II]	3587/3679, 6164, 8581, 9126, 14.4 μm, 33 μm
cl_3 [Cl III]	3344/3354, 5519, 5539, 8480 (multiplet)
cl_4 [Cl IV]	3120/3205, 5325, 7263, 7533, 11.8 μm, 20.3 μm
ca_2 Ca II	3935, 3970, 7293/7326, 8500/8544/8665
ca_5 [Ca V]	3999, 5311, 6088, 3.05 μm, 4.16 μm, 11.5 μm

Some registry ions are tracked in the ionization balance but carry no line block because CHIANTI v11 has no level/collision data for them: Ar II, Cl V, Ca III, Ca IV.

3.11 Free-free Gaunt factors and continuum data

gaunt.f90 is the verbatim MOCASSIN copy of the Hummer (1988) two-dimensional Chebyshev fit of the Karzas & Latter (1961) free-free Gaunt factor ($\leq 0.7\%$; interface getGauntFF(z, log10Te, xlf, g)). The thermal loop uses the thermally averaged $\bar{g}(T,Z) = \int g_{\rm ff}(u)\, e^{-u}\, du$ tabulated at setup ($\bar{g}(10^4\,{\rm K}, Z{=}1) = 1.2626$), now over net charge

$Z = 1$–4 so the trace-metal free-free (`par%metal_freefree`) has its Gaunt factor. The van Hoof et al. (2014) table is now available as a drop-in (`par%gaunt_vh14`; `data/gauntff_vh14.dat`, an 81×146 grid in $\log_{10}\gamma^2 = \log_{10}(Z^2\,\mathrm{Ry}/kT)$ and $\log_{10} u = \log_{10}(h\nu/kT)$, selected at both call sites — the cooling average and the continuum $\gamma_{\rm ff}$). The two compilations agree: switching moves T_e by a median 5×10^{-5}, at most 0.5%, consistent with the sub-percent Hummer accuracy van Hoof et al. report. Hummer remains the default so the recorded gates reproduce.

The nebular continuum tables `data/gamma{HI,HeI,HeII}.dat` (MOCASSIN inheritance; 21 temperatures × paired-edge frequency points) were *numerically verified* (2026-07-12) against a hydrogenic Kramers Milne sum: they contain free-bound *plus* free-free combined coefficients (at $\nu = 0.0069$ Ryd, 10^4 K the table value 9.47 decomposes into ff $\simeq$ 8.9 + fb $\simeq$ 0.6 in 10^{-40} units; near the Balmer edge the residual is $\sim$10%, the size of the neglected bound-free Gaunt factor). Consequently the port never adds free-free inside the table range. The Ercolano & Storey (2006) item is resolved as *provenance*: the MOCASSIN 2.02 tables ship in the ES06 compact paired-edge format, so no swap is needed — the verification above is the record. Two-photon continua follow MOCASSIN: Nussbaumer & Schmutz (1984) spectral shape with Pengelly (1964) $\alpha_{\rm eff}(2s)$ and the Osterbrock $q(2^2\mathrm{S}$–$2^2\mathrm{P})$ collisional suppression for H I; the Z-scaled form with Storey & Hummer $\alpha_{\rm eff}$ for He II; Almog & Netzer (1989) shape (`data/HeI2phot.dat`) with Benjamin, Skillman & Smits (1999) $\alpha_{\rm eff}$ for He I.

3.12 H I and He II recombination lines (Storey & Hummer 1995)

`make_sh95_lines.py` parses the SH95 case-B data (`e1bx.d`) with the reference reader indexing $k = (2n_{\rm top} - n_{ll} - n_l + 1)(n_l - n_{ll})/2 + n_{\rm top} - n_u + 1$ and exports Hα, Hβ, Hγ, Hδ, Pα, Pβ, Brγ on the full (T, n_e) grid (11 × 13) to `data/atomic/sh95_hi_caseB.txt`; Fortran interpolates bilinearly in $(\log T, \log n_e)$. Checks: $4\pi j_{\mathrm{H}\beta}/(n_e n_p) = 1.235\times10^{-25}$ at $(10^4\,\mathrm{K}, 10^2\,\mathrm{cm}^{-3})$ and Hα/Hβ = 2.864 (canonical values).

He II ($Z = 2$) comes from the *original* CDS file `e2b.d`, whose format differs from the repackaged `e1bx.d` (six-token block headers; index $k = (n_{\rm cut} - n_u)(n_{\rm cut} + n_u - 1)/2 + n_l$ with $n_{\rm cut} = 25$, from the SH95 `intrat.f` reader): six lines (1640, 4687, 3204, 10126, 5413, 4543 Å) on a 12×13 grid in `data/atomic/sh95_heii_caseB.txt`, weighted in the line table by $n_e n_{\rm HeIII}$. Checks: $4\pi j_{4687}/(n_e n_{\rm HeIII}) = 1.495\times10^{-24}$ at $(10^4\,\mathrm{K}, 10^4\,\mathrm{cm}^{-3}) = \alpha^{\rm eff}_{4687}(3.57\times10^{-13})\,h\nu$, and $1640/4687 = 6.56$ there.

He I lines use the same interpolator (a third table in `sh95_mod`) but a different source: the Porter et al. (2012, 2013) case-B emissivities, parsed by `make_hei_lines.py` from Cloudy c23.01's `he1_case_b.dat` (copied into `data/atomic`) onto a 21×14 (T, n_e) grid. Unlike the H I and He II tables, these are case-B collisional-radiative TOTALS, not pure recombination cascades: the tabulated $4\pi j/(n_e n_{\mathrm{He}^+})$ is density-dependent (the 10833 Å coefficient rises 6.6x from $n_e = 10$ to 10^4 cm^{-3} at 10^4 K) because it already includes collisional excitation out of the 2^3S metastable. Eight H II-region diagnostics are exported (3890, 4027, 4473, 5877, 6680, 7067, 7283, 10833 Å, vacuum), weighted in the line table by $n_e n_{\rm HeII}$ (He^+ recombining). Emergent ratios are physical: $5877/4473 = 2.88$, $6680/4473 = 0.82$, $7067/4473 = 0.49$ at $\sim$7 kK.

The H I and He II tables follow `par%case_ab`: case-A tables (`sh95_{hi,heii}_caseA.txt`, parsed from the original CDS files `e1a.d`/`e2a.d` onto the 10×9/12×9 grids case A tabulates) are loaded when the ionization runs case A, so the line emissivities match the balance; case B (Hα/Hβ = 2.94) is the default. The He I Porter set is case B only. An optional collisional H I Balmer component (`par%h_coll_effects`, off by default) solves a 25-level H I atom (from CHIANTI `h_1` through the n-level pipeline) with collisional excitation from the ground level, groups the l-resolved emission into Hα/Hβ/Hγ, weights it by $n_{\rm HI}$, and writes it as separate `hc` rows — a first-cut estimate (optically-thin cascade) for the front and hot-cell bias, kept apart from the recombination lines.

3.13 He I 2^3S metastable diagnostic

The Porter case-B 10833 Å table above is a collisional-radiative *total* emissivity, correct in the dense, optically-thin, recombination-dominated H II regime it was built for. It does *not* give the 2^3S population n_3, its line-of-sight column, or the 10833 Å *absorption* opacity — the observables of exoplanet transit, chromospheric, and PDR 10833 work, where the line is seen against a background. `hei_metastable_mod` (`par%hei_metastable`) adds these as a separate diagnostic; it does not add or modify any emission row, and in particular *nothing is summed onto* the Porter 10833 line (which already carries the metastable-collisional term — summing an explicit collisional 10833 would double-count).

The 2^3S balance (Oklopcic & Hirata 2018). OH18 track the neutral singlet ground 1^1S, the metastable triplet 2^3S, and the ion. Because $n_3/n_{\rm He} \sim 10^{-7}$ the balance is linear in n_3 — a closed-form quotient in each leaf, no iteration:

$$n_3 = \frac{\alpha_3\, n_e\, n_{\rm HeII} + q_{13}\, n_e\, n_{\rm HeI}}{A_{31} + \Phi_3 + n_e\,(q_{31a} + q_{31b}) + n_{\rm HI}\, Q_{31}}, \tag{18}$$

with source terms recombination into the triplet system ($\alpha_3\, n_e\, n_{\rm HeII}$) and e-impact excitation $1^1{\rm S} \to 2^3{\rm S}$ ($q_{13}\, n_e\, n_{\rm HeI}$), and sinks the radiative decay A_{31} ($2^3{\rm S} \to 1^1{\rm S}$), photoionization Φ_3, spin-changing e-impact transfer into the singlets ($q_{31a} = 2^3{\rm S} \to 2^1{\rm S}$, $q_{31b} = 2^3{\rm S} \to 2^1{\rm P}$ — *not* $2^3{\rm S} \to 1^1{\rm S}$), and Penning/associative ionization with neutral hydrogen ($n_{\rm HI}\, Q_{31}$). Two limits anchor the gates: the low-density radiative limit $n_3/n_{\rm He^+} = \alpha_3 n_e / A_{31}$ and the high-density collisional plateau $\alpha_3/(q_{31a} + q_{31b})$, crossing at $n_{\rm crit} = A_{31}/(q_{31a} + q_{31b})$.

The rate coefficients are closed-form fits transcribed from the EXHALE triplet implementation (offline into the coefficient file, Fortran only evaluates): $\alpha_3(T) = 2.10\times10^{-13}(T/10^4)^{-0.778}$ (OH18); q_{13}, q_{31a}, q_{31b} from Bray et al. (2000) collision strengths via OH18; $A_{31} = 1.272\times10^{-4}$ s^{-1} (the CHIANTI `he_1.wgfa` $A(1\text{–}2) = 1.73\times10^{-4}$ is $\sim$36% high and is *not* used); Q_{31} from Taylor et al. (2025) Table 2 (Garcia Muñoz 2025 recorded as the alternative, 1.2–2.2× lower — the documented uncertainty of the neutral/PDR sink). A consistency check: at 10^4 K, $\alpha_3 = 0.998 \times 0.75\, \alpha_B$(He II), matching the `HEI_F3S` = 0.75 triplet branching the diffuse field already uses.

The 2^3S photoionization rate. $\Phi_3 = \sum_\nu 4\pi J_\nu\, \sigma_3(\nu)\, \Delta\nu/(h\nu)$ over the *full* band (the reduced tally `jt_ion`, the same integral pattern as Γ_i). The threshold is 4.78 eV, but $\sigma_3 \sim 1\text{–}2\times10^{-18}$ cm^2 remains significant far above 13.6 eV (Cooper minimum near 34.7 eV, bump near 45.6 eV), so the ionizing band alone gives $\Phi_3 > 0$ wherever a field exists; `add_fuv` with `efuv_min`= 4.78 adds the soft-UV part [4.78, 13.6) eV (a rank-0 note fires when the band floor sits above 4.78 eV). $\sigma_3(E)$ is two VFKY96 wings joined by a log-linear bridge, a $\sim$4% fit to Norcross (1971) + TOPbase.

10833 Å absorption (Stage 2). From n_3 the line-center opacity of each fine-structure component is

$$\kappa_{0,i} = \frac{\sqrt{\pi}\, e^2}{m_e c}\, f_i\, \frac{\lambda_i}{b}\, n_3, \qquad b = \sqrt{2kT_e/m_{\rm He} + v_{\rm turb}^2}, \tag{19}$$

the classical integrated cross section $\pi e^2/(m_e c) = 0.02654$ cm^2 Hz divided by $\sqrt{\pi}\, b$ for the Gaussian peak. The three $2^3{\rm S} \to 2^3{\rm P}_J$ components (vacuum 10832.06 / 10833.22 / 10833.31 Å; f = 0.0599/0.1797/0.2996, ratio 1:3:5; A =

1.0216×10^{7} s^{-1} each) take the NIST ASD values (Drake 2007), cross-checked against p-winds; the CHIANTI `he_1.wgfa` $A \simeq 1.15\times10^{7}$ ($\sim$13% high) is *not* used. The $J' = 1,2$ pair sits 0.089 Å apart, far inside the $\sim$0.4 Å thermal Doppler width at 10^4 K, so they blend into the observed "doublet" (their line-center opacities are summed); $J' = 0$ at 10832.06 Å is resolved. The Python `HeiMetaData` reader deposits n_3 and κ_0 along the line of sight (the flux-conserving deposit of the emissivity maps) into column [cm^{-2}] and $\tau_0 = \int \kappa_0\, \mathrm{d}l$ maps, and warns when line-center $\tau_0 > 1$.
The module runs once at output time (no hot-loop cost, no feedback — n_3 is a $\sim10^{-7}$ trace of He). Caveats recorded at the code site: (i) *local equilibrium* — the advection term $v\,\mathrm{d}f_3/\mathrm{d}r$ is absent, valid where the 2^3S destruction time is short against any flow time (dense H II), not a faithful outflow benchmark; (ii) 10833 can be optically thick in dense metastable columns, where the optically-thin τ_0 diagnostic breaks down (line transfer is out of scope).

4. Algorithms

4.1 The iteration loop

read grid $\rightarrow$ persist gas leaf state $\rightarrow$ fill `kap_ion`
do iter = 1, `gas_niter`
  zero the band tally J_b
  sample and transport stellar packets (analytic edge walk, Eq. 4)
  [diffuse] rebuild the diffuse emission table from the current state;
    sample and transport $L_{\mathrm{dif}}/L_{\mathrm{pkt}}$ diffuse LyC packets
  ALLREDUCE J_b; rate integrals (Eq. 5); metal Γ's
  solve each leaf: ionization only or coupled (x, T_e)
  refill `kap_ion` from the new state (the stellar tally is
    recomputed from zero each iteration — no reuse)
  converged by `par%conv_crit`: cell-max or volume-integrated tests (below)
end do
outputs: rates + state maps; line luminosities; nebular continuum

Every rank runs the identical cell solves (inputs are the reduced tallies), so all ranks agree on convergence; only the node master writes the shared state arrays. Under-relaxation on x (`par%ion_relax`) is available for sharp I-fronts. Practical lessons recorded: a fully neutral start advances the front only $\sim$one cell per iteration (the photons die in the first neutral cell), while an ionized start converges globally; and $\max|\Delta x_{\mathrm{HII}}|$ stalls at the front-cell Monte-Carlo noise floor ($\sim$2–4×10^{-2} at 4×10^{7} packets), so tolerances below that are unreachable by that criterion — the diffuse field, interestingly, smooths the front and converges to 3×10^{-3}.

4.2 Stellar packet launch

For each direct packet, MoCHII first samples its energy with Eq. (1); it then samples an isotropic point-source direction with $\mu = 2u_\mu - 1$ and $\phi = 2\pi u_\phi$. The source position, the exact energy, the direction, and the common packet luminosity are placed in the packet before the octree edge walk begins. Thus energy sampling is not conditional on the diagnostic binning. At every traversed cell the path estimator uses the packet's exact-energy opacity and cross sections; the energy-bin index is accumulated only alongside this estimator to form the reported spectrum. Rebuilding the direct tally from zero on every nonlinear iteration means that the changing opacity is always applied to a new sample of the source spectrum rather than to a reweighted old tally.

For ordinary pseudo-random launching the inverse-CDF variate and the two direction variates are independent Mersenne-Twister draws. The same probability law is used by the Sobol option described next. The packet weight is not divided by the probability density: equal packet luminosity combined with inverse-CDF sampling is the importance sampling rule that recovers the source luminosity spectrum in expectation.

Convergence measures. Two pairs are computed every iteration and `par%conv_crit` selects which gates the loop: the cell-max tests ($\max|\Delta x_{\rm HII}| < \texttt{gas\_tol}$, $\max|\Delta T_e|/T_e < \texttt{gas\_tol\_te}$) and the volume-integrated tests

$$\frac{\sum_{\rm leaf} V\,|\Delta x_{\rm HII}|}{\sum_{\rm leaf} V\,x_{\rm HII}} < \texttt{gas\_tol}, \qquad \frac{\sum_{\rm leaf} V\,n_e\,|\Delta T_e|}{\sum_{\rm leaf} V\,n_e\,T_e} < \texttt{gas\_tol\_te}. \tag{20}$$

The volume measures fix both stall modes of the cell-max tests: front jitter occupies little volume (the front shell is a few thousand cells against $\sim 10^5$ ionized ones, so its noise enters the L_1 norm suppressed by that ratio), and the n_e weight removes the vacuum and no-heating cells whose `te_min`-pinned oscillation dominates $\max|\Delta T_e|$ in FUV/PDR configurations. `'vol'` is the recommended production setting; `'cell'` (the historical behavior) remains the default so the recorded gates reproduce. Measured on the FUV dusty Strömgren case at 8×10^6 packets (where the cell-max tests sat at 0.1 and 6.4 indefinitely): the volume measures decay smoothly to well-defined Monte Carlo floors of 2.7×10^{-3} (x) and 6.6×10^{-3} (T_e) by iteration $\sim$27, and running 30 further iterations leaves the solution unchanged ($V_{\rm ion}$ to 0.014%) — the floor IS convergence. Set the tolerances just above the floor for the packet count in use (here 5×10^{-3} and 10^{-2} would stop the loop at iteration $\sim$23); the floors scale as $1/\sqrt{N_{\rm packets}}$. With `par%require_convergence` (default off) a non-converged gas iteration prints a warning and, when the switch is set, aborts before writing output; either way the rates-file header records the convergence status in the keywords `CONVERGD`, `NITERDON`, `FINALDX`, and `FINALDTE`.

4.3 Quasi-random photon launching (optional)

With `par%launch_sequence = 'sobol'` the packet launch draws its variables — source component where relevant, energy, direction, and any entry-surface coordinates — from an Owen-scrambled Sobol net (`src/qmc_mod.f90`) instead of the Mersenne Twister, while every post-launch transport event stays pseudo-random. The Sobol point is addressed by the *global* photon number, so the launch set is independent of the MPI task count, and `par%qmc_seed` is held fixed across the nonlinear iterations (common random numbers). Because the band has no scattering, the direct field is an analytic function of the launch variables (Eq. 4), so a low-discrepancy launch set reduces its noise directly: on the fixed-state analytic-attenuation gate the cell-wise $\Gamma_{\rm HI}$ error scales as $\sim 1/N$ (Monte Carlo $1/\sqrt{N}$), with effective photon gains of 29× at $N = 2^{17}$ and 222× at 2^{20}, at unchanged wall time and no significant bias. The method, the fixed dimension layouts, and all gates are documented in the companion memo `MoCHII_QMC`. In the validated single point-source continuous mode the three Sobol coordinates are (u_E, u_μ, u_ϕ): u_E is inverted through the source CDF and the remaining coordinates set the isotropic direction. QMC therefore reduces the discrepancy of the energy samples as well as of the angular samples; it does not replace continuous energy sampling by bin selection. Diffuse packets use a separate scrambled stream and the nine coordinates listed in §4.8.

4.4 Ionization equilibrium

Per leaf at temperature T (`solve_ion_cell`): with $R_i = \Gamma_i + n_e C_i(T)$ the ionization rates and α_i the case-A/B recombination rates, $x_{\rm HI} = \alpha n_e/(R_{\rm HI}+\alpha n_e)$ and the He stage chain $n_{i+1}/n_i = R_i/(\alpha_{i+1} n_e)$, closed by $n_e = n_{\rm H}[x_{\rm HII} + y_{\rm He}(x_{\rm HeII} + 2x_{\rm HeIII})]$ and solved by a damped fixed point on n_e (mixing $\frac{1}{2}$, converged at 10^{-10}, ≤ 200 iterations).

Two-way charge exchange (`par%charge_exchange = 'two_way'`, the default). Each reaction $X^{i+} + H^+ \leftrightarrow X^{(i+1)+} + H^0$ (§3.8) moves one unit of charge between a metal and hydrogen, so hydrogen's balance carries it too:

$$n_{\rm HI}\,(\Gamma_{\rm H} + n_e C_{\rm H} + R_{\rm cx}^{\rm out}) \;=\; (\alpha_{\rm H} n_e + R_{\rm cx}^{\rm in})\, n_{\rm HII}, \tag{21}$$

with the sums running over every registry element and transition,

$$R_{\rm cx}^{\rm in} = \sum_{\rm el}\sum_i k_{{\rm ion},i}^{\rm CX}\, n(X^{i+}), \qquad R_{\rm cx}^{\rm out} = \sum_{\rm el}\sum_i k_{{\rm rec},i}^{\rm CX}\, n(X^{(i+1)+}), \tag{22}$$

in s^{-1}: $R_{\rm cx}^{\rm in}$ neutralizes an H^+, $R_{\rm cx}^{\rm out}$ ionizes an H^0. Eliminating $n_{\rm HII} = n_{\rm H}(1 - x_{\rm HI})$ keeps the closed form,

$$x_{\rm HI} = \frac{\alpha_{\rm H} n_e + R_{\rm cx}^{\rm in}}{\Gamma_{\rm H} + (C_{\rm H} + \alpha_{\rm H})\, n_e + R_{\rm cx}^{\rm in} + R_{\rm cx}^{\rm out}}, \tag{23}$$

so no matrix and no Newton step are needed; *both* charge-exchange terms appear in the denominator, $R_{\rm cx}^{\rm in}$ for the same reason $\alpha_{\rm H} n_e$ does. The two sums come from the same walk of the stage chain that produces the metal electrons (`species_metal_sums`) and read the same cached coefficients the cascade reads, so the charge each reaction moves is conserved in the reacting pair by construction rather than to a tolerance; the gate `tests/charge_exchange/` forms all 29 active reactions from both sides and finds them equal to 6×10^{-16} relative. The stage densities are taken from the previous iterate of the n_e fixed point, since they depend on $n_{\rm HI}$ and $n_{\rm HII}$ themselves. The value `'metal_only'` keeps the charge exchange in the metal cascade alone — the earlier bookkeeping, which does not conserve charge — and is retained only to reproduce the recorded gates.

The correction is small on these benchmarks, and the reason is worth stating because it is not the usual abundance argument. Where the near-resonant $O^0 + H^+$ pair locks oxygen to hydrogen — that is, where $\Gamma_{\rm O}$ and $n_e\alpha_{\rm O}$ are negligible beside the charge-exchange terms — detailed balance of the pair gives $R_{\rm out}/R_{\rm in} = n_{\rm HII}/n_{\rm HI}$, and substituting that into Eq. (23) returns $x_{\rm HI} = \alpha_H n_e/[\Gamma_H + (C_H + \alpha_H) n_e]$ exactly: the two terms cancel identically. What survives is only the departure from that lock. Measured with one switch changed and the same Sobol seed, $\langle T[N_{\rm p}N_{\rm e}]\rangle$ moves by −1.41 K at HII40 and −0.13 K at HII20, $L({\rm H}\beta)$ by +0.045% and +0.007%, and $r({\rm He}^0 = 0.5)$ by −0.0002 and −0.0000 pc. All of these are inside the tolerance the runs were converged to (`par%gas_tol_te` = 10^{-2}, i.e. 82 K at HII40) by factors of 58 and 533, so the recorded benchmark tables are not re-baselined on them.

The same cancellation has a numerical consequence, and it dictates how the balance is solved. Substituting into Eq. (23) with the charge-exchange terms lagged gives a map whose slope at the fixed point is $R_{\rm in}/(R_{\rm in} + \alpha_H n_e)$, which tends to unity wherever $R_{\rm in} \gg \alpha_H n_e$ — that is, throughout partially neutral gas, where the iteration becomes marginally stable. At $T = 10^4$ K, $n_{\rm H} = 100$ and the HII20 abundances it then needs 124 passes at $x_{\rm HII} = 0.18$, 239 at 0.10 and 1285 at 0.021, against a 200-pass cap, and a capped cell leaves with $x_{\rm HII}$ high by about 8%.

MoCHII therefore does not substitute. `hydrogen_neutral_fraction` solves the balance at the current n_e for its root, with the whole metal cascade re-evaluated at each trial $x_{\rm HI}$. The root is bracketed in $[0,1]$ *by construction* — $R(0) = -(\alpha_H n_e + R_{\rm in}) < 0$ and $R(1) = \Gamma_H + (C_H + \alpha_H) n_e + R_{\rm out} > 0$ — so Illinois iteration keeps the bracket at every step and the fraction cannot leave the physical range, which is the one property the plain substitution had and a bare secant would forfeit. One call now reaches the converged state to 3.6×10^{-11} at every $x_{\rm HII}$ down to 0.02, against 1.8×10^{-3} for substitution, and on HII20 the printed balance residual falls from a maximum of 4.6×10^{-3} to 1.3×10^{-6} with the volume-weighted mean from 4.7×10^{-4} to 4.5×10^{-9}. The benchmark numbers above move by 0.01 K, so the method changed the path to the root and not the root. Extrapolating the n_e sequence instead — Aitken — was tried and made the pass count worse, because a jump in one component leaves the rest of the coupled state off the slow manifold; that is recorded with its measurements in `docs/ION_BALANCE_CONVERGENCE_PLAN.md`.

Because the written state is under-relaxed (`par%ion_relax`) while the metal stage fractions written alongside are evaluated at the written n_e and T_e, the residual of Eq. (21) on that state is not identically zero; `hydrogen_balance_residual` reports it each gas iteration as the maximum over leaves and the volume-weighted mean. It is a printed diagnostic and does not gate convergence.

4.5 Thermal balance

When `par%solve_te` is set, `gas_thermal_update` replaces the fixed-T `gas_equilibrium_update` (§4.4) and updates each leaf's temperature once per iteration, from that iteration's radiation field, by finding the equilibrium $\mathcal{H}(T) = \mathcal{C}(T)$ — the root of the net rate

$$\mathcal{N}(T) = \mathcal{H}(T) - \mathcal{C}(T), \qquad \mathcal{H} = n_{\rm HI}\mathcal{H}_{\rm HI} + y_{\rm He} n_{\rm H}(x_{\rm HeI}\mathcal{H}_{\rm HeI} + x_{\rm HeII}\mathcal{H}_{\rm HeII}), \tag{24}$$

with the photoheating rates $\mathcal{H}_i$ (per atom) from Eq. (5). The ionization state is *re-solved* (`solve_ion_cell`, §4.4) at every trial T: both $\mathcal{H}$ and $\mathcal{C}$ depend on the fractions $(x_{\rm HI}, x_{\rm HeI}, x_{\rm HeII}, n_e)$, which themselves depend on T. The ionization response is mild, so $\mathcal{N}(T)$ is monotone through its root and bisection on $\log T_e$ in [`te_min`, `te_max`] is safe: each step takes the geometric midpoint $\sqrt{T_{\rm lo} T_{\rm hi}}$ (up to 60 steps, stopping at $T_{\rm hi}/T_{\rm lo} - 1 < 10^{-5}$), and if cooling already wins at `te_min` ($\mathcal{N} \le 0$) the leaf is pinned to `te_min` while if heating still wins at `te_max` it is pinned to `te_max`.

The cooling $\mathcal{C}(T)$ (`cooling_mod`) assembles recombination (Hui–Gnedin β), free-free ($1.42554 \times 10^{-27} \sqrt{T} Z^2 \bar{g} n_e n_{\rm ion}$, summed over H II, He II ($Z = 1$) and He III ($Z = 2$) with Z the *net* ion charge; under `par%metal_freefree` (on by default) the trace metals are added with $Z_{\rm ion}$ = (stage −1), not the nuclear Z, using $\bar{g}(T, Z_{\rm ion})$ tabulated for Z = 1–4 — singly-ionized metals dominate this small $\sim 10^{-4}$-of-the-H/He term), collisional ionization ($k_{\rm ci} \times$ ionization potential), H I line cooling, and the trace-metal sum $\sum n_e n_{\rm ion} \Lambda_{\rm ion}(T, n_e)$ with the cascade fractions (§4.6) re-evaluated at each trial T. The line-cooling coefficient $\Lambda_{\rm ion}(T, n_e)$ is the density-suppressed value of Eq. (16) by default (`par%cooling_model = local_ne`), the plain fit $\Lambda_{\rm ion}(T)$ under `low_density`. PDR runs add three optional terms to $\mathcal{N}$: metal photoheating (`par%metal_heat`); grain photoelectric heating with grain recombination cooling (Bakes & Tielens 1994, `par%grain_pe`, driven by the local FUV field G_0, both weighted by $x_{\rm HI}$ and evaluated at $\min(T, 10^4\,{\rm K})$ so the fits stay in range — two guards the PDR gate needed against a 12–35 kK runaway); and the H-impact [C II] 158 μm / [O I] 63 μm fine-structure cooling that balances the photoelectric heating in the mostly neutral gas.

Under `par%use_sec_ion` (Shull & van Steenberg 1985) a photoelectron of kinetic energy $E_0 = h\nu - E_{{\rm th},s}$ from absorber

s (H I, He I, He II) with $E_0 > 40$ eV deposits only $f_{\rm heat}(x)$ of its excess as heat (below the threshold it thermalizes fully), so the photoheating integrand is weighted by $f_{\rm heat}(x)$; the un-thermalized balance drives $f_{\rm ion,HI}(x)E_0/E_{\rm th,HI}$ secondary H I and $f_{\rm ion,HeI}(x)E_0/E_{\rm th,HeI}$ secondary He I ionizations that add to the corresponding photoionization rates in `solve_ion_cell`. Here $x = (n_{\rm HII}+n_{\rm HeII}+n_{\rm HeIII})/(n_{\rm H}+n_{\rm He})$ is the ionized fraction of the H+He nuclei and the partitions are $f_{\rm heat} = 0.9971[1-(1-x^{0.2663})^{1.3163}]$, $f_{\rm ion,HI} = 0.3908(1-x^{0.4092})^{1.7592}$, $f_{\rm ion,HeI} = 0.0554(1-x^{0.4614})^{1.6660}$. The effect vanishes in fully ionized gas ($f_{\rm ion} \to 0$ as $x \to 1$, so the HII-region gates are unchanged) and matters only in partially neutral gas under a hard field (fronts, PDR, X-ray/AGN); metals are neglected as absorbers (trace). Default is off.

The solved T_e is written directly (only the ionization fractions carry the under-relaxation `par%ion_relax`), n_e is rebuilt from the new state — including the trace-metal electrons under `par%metal_ne` — and, as in the ionization-only path, only the node-local rank 0 runs the bisection (the reduced band tally is identical on every node master) and writes the shared state, with the convergence metrics broadcast within the node. The loop is gated by the T_e convergence measures defined above (`gas_tol_te`, cell-max or volume-integrated per `par%conv_crit`).

4.6 The trace-metal cascade

The registry (`species_mod`) activates an element when its abundance is positive. Stage fractions follow the product chain

$$\frac{n_{i+1}}{n_i} = \frac{\Gamma_i + n_e C_i + n_{\rm HII}\, k^{\rm CX}_{{\rm ion},i}}{n_e\,[\alpha_{\rm RR}+\alpha_{\rm DR}]_{i+1} + n_{\rm HI}\, k^{\rm CX}_{{\rm rec},i+1}}, \tag{25}$$

with metal Γ_i computed once per iteration from the same band tally. The chain is evaluated on a known H/He and thermal state rather than solved jointly with it, but the metals are not excluded from the ionizing opacity (§4.7), from n_e (`par%metal_ne`), or from hydrogen's own balance: the two $k^{\rm CX}$ terms above enter Eq. (21) as well (§4.4). Adding an ion is one entry in the data generators plus its cooling and n-level files.

4.7 Metal opacity in the He-ionizing window

"Trace" names the *data* treatment, not a switched-off feedback. The founding plan lists metals feeding back on n_e and on the opacity as a non-goal, but the production inputs run with `par%metal_ne`, `par%ion_metal_abs` and `par%metal_heat` all true, so both feedbacks are active — and since 2026-07-31 the first and third are also the shipped defaults, for the reason measured below.

The two switches separate cleanly and add. Switching both off costs 21.3 K (0.26%) at HII40 and 5.8 K (0.084%) at HII20; `metal_heat` carries nearly all of it (–21.1 and –5.7 K) and `metal_ne` almost none (–0.2 and –0.1 K), with the fronts moving by at most 0.003 pc. The smallness of `metal_ne` is not an argument that metal electrons do not matter: the metal stages hold about 0.1% of the electrons in the ionized gas, and $\langle T[N_{\rm p}N_{\rm e}]\rangle$ is weighted by $N_{\rm p}N_{\rm e}$, which suppresses exactly the neutral gas where the metal stages are the *only* electron source and n_e would collapse without them. That is why the bound above does not transfer past the ionization front, and why both now default on: an approximation is worth keeping only where its cost is known, and beyond the front it is not.

Concretely, `species_opacity_add` sums over *all* active stages with their VFKY96 cross sections and the self-consistent stage fractions, and the sum is iterated together with the opacity it changes. The cross sections themselves are not

Table 3: What the two metal feedbacks are worth on the benchmarks. Each run changes only the named switch; same inputs, same Sobol seed. Fronts from the volume-weighted 0.01-pc shell diagnostic.

`metal_ne`	`metal_heat`	$\langle T[N_{\rm p}N_{\rm e}]\rangle$ [K]	$L(\mathrm{H}\beta)$	$r(\mathrm{He}^0{=}0.5)$	$r(\mathrm{H}^0{=}0.1)$
HII40					
on	on	8168.9	1.9293e37	4.1278	4.6111
on	off	8147.9	1.9306e37	4.1248	4.6090
off	on	8168.7	1.9291e37	4.1291	4.6123
off	off	8147.6	1.9304e37	4.1259	4.6100
HII20					
on	on	6924.5	4.6837e36	1.2408	2.8352
on	off	6918.8	4.6839e36	1.2406	2.8345
off	on	6924.4	4.6836e36	1.2408	2.8358
off	off	6918.7	4.6838e36	1.2407	2.8352

in question. For the eight ions that carry the He-ionizing window — C I–III, N II, O II, Ne I, S II, S III — MoCHII reproduces Verner's own `phfit2.f` to 10^{-6} relative, the single-precision round-off of `phfit2` itself.

What is genuinely absent is metal recombination continua as a diffuse *source*: the diffuse field has exactly four channels (`dif_ch(4, nleaf)`) and all four are H/He. Summing the metal ground-state recombination continua whose photons lie above 24.587 eV (C IV, N III, N IV, O III, O IV, Ne III, Ne IV, S III, S IV) gives 0.29% of the He I ground-continuum channel, about 0.001 pc of front position. Metals as secondary-ionization absorbers are absent as well, but `par%use_sec_ion` is off in the benchmarks.

The measured lever. Scaling the summed metal cross section on the converged HII40 model (68 ranks, same Sobol seed, 0.01-pc shell diagnostic) gives

variant	$r(\mathrm{He}^0{=}0.5)$ [pc]	Δr	pc per unit fraction
production	4.11970	—	—
metal $\sigma \times 0.70$	4.04764	−0.07206	0.240
metal $\sigma \times 1.30$	4.15063	+0.03093	0.103
metal absorption off	3.50470	−0.61500	0.615
metals off entirely	1.93335	−2.18635	(also removes all metal cooling; T_e 8162 $\to$ 18681 K)

The response is strongly asymmetric and saturating: removing metal opacity costs 0.615 pc per unit while adding it gains only 0.103 pc per unit. Extrapolating the upward branch, closing the 0.178 pc Cloudy residual of §4.8 would take +180% metal opacity ($\times 2.8$), and the saturation means still more. That is not available, and it retires the metal opacity as a single-parameter explanation.

The prediction recorded before these runs was "metal opacity $\times 1.30$ moves $r(\mathrm{He}^0{=}0.5)$ outward, $\times 0.70$ inward, of order 0.1–0.2 pc". The *sign* was right both ways; the *magnitude* was wrong by factors of 1.5–6, because the prediction extrapolated the on/off bound of 0.615 pc as if the response were symmetric. An on/off bound measures what removal costs and says little about what addition could gain. The companion statement that "a $\sim$30% error in the metal opacity would be worth $\sim$0.18 pc" is accordingly withdrawn: 30% is worth 0.031–0.072 pc.

Mechanism: spectral hardening, not cooling. The metal stages with thresholds inside 21–40 eV (Ne I 21.56, S II 23.33, C II 24.38, N II 29.60, S III 34.83, O II 35.12 eV) preferentially remove the *softest* He-ionizing photons, and $\sigma(\mathrm{He\,I})/\sigma(\mathrm{H\,I})$ rises steeply with energy — 6.1 at 25 eV, 9.0 at 35, 13.9 at 55, 19.9 at 95 eV. The hardened surviving field is therefore taken up by He^0 rather than H^0 and the He^+ zone extends; remove the metal opacity and the field stays soft, H^0 takes a larger share, and the zone collapses. The alternative readings are excluded by the same runs: between production and `ion_metal_abs = .false.` the hydrogen structure and the mean temperature barely move ($r(\mathrm{H}^0{=}0.1) = 4.61633$ vs. 4.61551 pc; T_e 8162 vs. 8183 K), so the 0.615 pc is neither a thermal effect nor a change in the hydrogen solution.

What Cloudy has here that MoCHII does not. Cloudy's base is the same Verner `phfit`. Its `opacity_createall.cpp` then adds channels MoCHII has no equivalent for: N^+ excited-level photoionization (Henry 1970), O I valence replaced wholesale by `ofit` when PHFIT96 is in use, O I ^{1}S, O^{2+} ^{1}D and ^{1}S, photoionization into excited states of O^+, and a Mg^+ excited level, several of them fitted to Opacity Project data. So Cloudy photoionizes metals out of excited levels and MoCHII only out of the ground level. As opacity these are small: the excited-level populations are 10^{-5}–10^{-4} of the ground, so the O^{2+} ^{1}D opacity is of order 10^{-4} of the O^+ ground opacity.
An independent Opacity Project comparison is possible for C, N and O through the TOPbase tables carried by `sirocco`. Photon-weighted with a 40-kK blackbody over 24.587–54.4 eV, OP/VFKY96 is 0.946 (C II), 0.978 (N II) and 0.863 (O II) — OP is 2–14% *lower*. With the measured upward derivative that is −0.005 to −0.03 pc, the wrong direction for the residual. Ne and S are absent from that table and were not checked.

A justification that had to be retracted in the code. Two comments — at the `ion_metal_abs` declaration in `define.f90` and above `species_opacity_add` — used to justify the metal opacity as "negligible next to H/He above 13.6 eV". The measurement above contradicts that, and both now carry the measured derivatives instead: the opacity is small in *magnitude* but not in *effect*, because it acts on the spectral shape in the band where He^0 and H^0 compete. An on/off statement is the form such a justification has to take here, since the response is asymmetric and a single number in either direction misstates it.

4.8 Explicit diffuse LyC field and packet sampling

Case B on the spot is replaced by case A rates plus explicit emission of ground-recombination packets. For each leaf and each of the three ground continua H II→H I($1s$), He II→He I(1^1S), and He III→He II($1s$), the emissivity uses $\alpha_1(T) = \alpha_A(T) - \alpha_B(T)$. At the start of every gas iteration the code rebuilds the energy-luminosity table

$$L_{\mathrm{ch},l} = n_{e,l} n_{\mathrm{rec},l} \alpha_{1,\mathrm{ch}}(T_l) \langle E \rangle_{n,\mathrm{ch}}(T_l) V_l, \qquad L_{\mathrm{dif}} = \sum_{l,\mathrm{ch}} L_{\mathrm{ch},l}, \tag{26}$$

with $\langle E \rangle_n$ the photon-number-weighted mean energy of the emitted continuum.

Ground-continuum spectrum (`par%diffuse_energy_model`). Detailed balance between photoionization and radiative recombination fixes the free–bound spectrum. Per unit photon energy the photon-number emissivity of each ground continuum is

$$n(E) \propto E^2\,\sigma_{\rm pi}(E)\exp\left[-\frac{E-E_{\rm th}}{kT_e}\right], \qquad E \geq E_{\rm th}, \tag{27}$$

evaluated with the *same* ground-state cross section the transport absorbs with, so a packet is absorbed with the cross section it was emitted with. This is the default, `'milne'`. Restoring the Milne prefactor turns the same integral into the ground-state recombination coefficient,

$$\alpha_1(T) = \frac{g_n}{2g_+}\frac{2}{\sqrt{\pi}}(kT)^{-3/2}\sqrt{\frac{2}{m_e}}\frac{1}{m_e c^2}\int_{E_{\rm th}}^{\infty} E^2\sigma_{\rm pi}(E)e^{-(E-E_{\rm th})/kT}\,dE, \tag{28}$$

so the spectrum and the emission rate are not independent assumptions. Equation (28) reproduces $\alpha_{1s}(\text{H I}, 10^4\ \text{K}) = 1.579\times10^{-13}\ \text{cm}^3\ \text{s}^{-1}$ against the tabulated 1.58×10^{-13}. The code reports that integral at start-up against the $\alpha_A - \alpha_B$ that sets the packet count and rescales neither side. Since the correction described next, the largest disagreement over the three channels is 0.01%. It was 3.44% before, all of it in He I, and that number is what exposed the defect.

The α_1 source-mixing defect. `recomb_mod` took α_A from the Badnell total, α_1 from the *level-resolved* Mao & Kaastra fit, and then defined $\alpha_B = \alpha_A - \alpha_1$. That subtraction crosses two unrelated determinations. Nothing in the construction forces the Mao n=1 value to be the ground term of the Badnell total it is subtracted from, and whatever separates them lands entirely in α_B — and, through α_1, in the diffuse packet count as well.

The cross-section side is not the error. MoCHII's `sigma_HeI` reproduces Verner's own `phfit2.f` to a ratio of 1.00000 at every energy over 24.6–105 eV, and its threshold value, 7.436 Mb, sits 0.5% above the measured 7.40 ± 0.07 Mb of Samson et al. (1994), inside the experimental uncertainty. Nor is the start-up comparison between unlike quantities: because $\alpha_B = \alpha_A - \alpha_1$, the "rates" side of it *is* α_1(He I) itself, so the 3.44% separated two determinations of one number.

Cloudy settles which side is right. Cloudy c25.00 derives its H-like and He-like recombination level by level from its *own* photoionization cross sections through the exact Milne relation (`source/iso_radiative_recomb.cpp`), caching the result in `data/h_iso_recomb.dat` and `data/he_iso_recomb.dat`. In those files row 0 of each element is the ground level and the last row is the total — verified on hydrogen, where row 0 is 1.5840×10^{-13} and the last row 4.1642×10^{-13} against the known $\alpha_{1s} = 1.58\times10^{-13}$ and $\alpha_A = 4.18\times10^{-13}\ \text{cm}^3\ \text{s}^{-1}$. At 10^4 K, in $10^{-13}\ \text{cm}^3\ \text{s}^{-1}$:

	α_A		α_1		
channel	MoCHII	Cloudy	Mao & Kaastra	Milne integral	Cloudy
H I	4.1933	4.1642	1.5873	1.5790	1.5840
He I	4.3719	4.3583	**1.5650**	1.6086	1.6032
He II	21.895	21.817	6.5366	6.5116	6.5145

Badnell's totals are accurate to 0.3–0.7% on all three channels, and the Mao n=1 values to 0.2–0.3% for H I and He II — but He I is 2.38% low, and the subtraction propagates that into α_B(He I) as +1.88%. MoCHII's own Milne integral matches Cloudy to 0.3%, 0.34% and 0.04%. The Milne relation is exact, Cloudy applies it to a He^0 ground cross section that agrees with Verner's to the 0.4% implied by the α_1 agreement, and so *the Milne route is the correct one and the level-resolved fit is the low side*.

An earlier reading of the same numbers, withdrawn. This document previously concluded the opposite — that the Milne integral was the outlier — from the third determination available at the time, Benjamin, Skillman & Smits (1999) with J. S. Mathis as carried by K. Wood's `ionize`, $\alpha_1(\mathrm{He\,I}) = 1.54\times10^{-13}(T_4)^{-0.486}$ cm^3 s^{-1}:

T	BSS99 + JSM	Mao & Kaastra	Milne integral	Cloudy c25.00
5000 K	2.157	2.184 (+1.3%)	2.259 (+4.8%)	—
10^4 K	1.540	1.565 (+1.6%)	1.609 (+4.5%)	1.603 (+4.1%)
2×10^4 K	1.100	1.128 (+2.6%)	1.151 (+4.7%)	—

In 10^{-13} cm^3 s^{-1}, percentages against the BSS99 column. The arithmetic was right — the Milne integral does sit 4.5–4.8% above BSS99 and 2.8% above Mao while the two fits cluster within 2.6% of each other — but the inference was a majority vote taken on the mutual closeness of two fits, which is not evidence that either reproduces the cross sections in use. Cloudy, computing the same quantity level by level from its own cross sections, lands within 0.34% of the Milne integral. The earlier conclusion is withdrawn.

The correction. `tools/fitting/fit_alpha1_milne.py` evaluates Eq. (28) on the cross sections the transport actually uses — exact hydrogenic for H I and He II (`sigma_hydrogenic`, $Z = 1$ and 2; note that `sigma_HI` in `src/photo_xsec.f90` is the exact hydrogenic form, not VFKY96) and VFKY96 for He I — at 241 temperatures over 10^3–10^5 K, and refits the same seven coefficients of Eq. (8), so that no Fortran structure changes:

function	$(a_0, b_0, c_0, a_1, b_1, a_2, b_2)$
`alpha1_HII`	1.87007×10^{-2}, 1.29077, -2.17465×10^{-3}, 1.29189×10^{1}, 8.05060×10^{-1}, 8.44399×10^{-2}, 5.93940×10^{-4}
`alpha1_HeII`	3.26862×10^{-3}, 6.97361×10^{-1}, 3.14907×10^{-4}, 5.76341×10^{1}, 1.18534, 2.58417×10^{1}, 9.90134×10^{-1}
`alpha1_HeIII`	2.61525×10^{-1}, 1.39527, -3.22079×10^{-4}, 6.40708×10^{1}, 8.97582×10^{-1}, 5.01551×10^{-1}, 1.21533×10^{-3}

The largest $|\mathrm{fit/Milne}-1|$ is 0.019% (H I), 0.001% (He I) and 0.006% (He II) over 10^3–10^5 K, and 0.0075%, 0.0004%, 0.0021% over the production range 3×10^3–3×10^4 K, so the functional form is not the limiting approximation. At 10^4 K the fitted α_1 is 1.5791, 1.6086 and 6.5117×10^{-13} cm^3 s^{-1}: within 0.35% of Cloudy on all three channels, with He I corrected from −2.38% to +0.34%. The code's own cross-check confirms it, `milne_setup` reporting a largest α_1 disagreement of 3.44% before and 0.01% after.

The gate is `tests/milne/check_alpha1.f90`, wired into `tests/milne/run_check.sh`. It re-implements the Milne quadrature independently from `photo_xsec`, checks the three channels at 81 temperatures over 10^3–10^5 K to 10^{-3}, checks $\alpha_1 < \alpha_A$ everywhere (largest α_1/α_A 0.560, 0.496, 0.445), and checks all three against the hardcoded Cloudy c25.00 values to 1%. Reinstating the old Mao He I coefficients makes it fail on both counts (3.886×10^{-2} from the quadrature, 2.383% from Cloudy) and it passes again after reverting.

Effect on the benchmarks, graded against the prediction. What was recorded before the runs: "α_1(He I) rises 2.4%, so channel 2 emits more; α_B(He I) falls 1.9%, so there is less case-B recombination. Both push the front outward, toward

Cloudy, by of order 0.015–0.02 pc." Measured on the paired HII40/HII20 production runs (68 ranks, same Sobol seed, converged in 38 and 27 iterations as before):

	HII40	HII20
$\Delta r(\mathrm{He}^0{=}0.5)$	+0.00811 pc	+0.00210 pc
Cloudy residual	−0.18561 → −0.17750 pc	—
$\Delta L(\mathrm{H}\beta)$	−0.3713%	−0.4159%
mean $T_e(x_{\mathrm{HI}} < 0.95)$	9053.4 → 9051.7 K	8522.3 → 8528.7 K

The *sign* was right, outward and toward Cloudy on both benchmarks. The *magnitude* was about half the low end of the predicted range, +0.0081 against 0.015–0.02 pc, so the prediction was roughly a factor two high. `tests/continuous_energy/check_hii_production.py` passes on both, with energy closure 1.5×10^{-16} and 0. The published benchmark quantities move by less than the width of the published spread: $\langle T[N_pN_e]\rangle$ from 8162 to 8169 K at HII40 and 6917 to 6925 K at HII20, T_{inner} from 7535 to 7536 and 6936 to 6938 K, and R_{out} from 8.89 to 8.88×10^{18} cm at HII20 with 1.44×10^{19} cm unchanged at HII40.

$L(\mathrm{H}\beta)$ moves the *wrong* way. It falls 0.37–0.42% ($1.94 \rightarrow 1.93\times10^{37}$ and $4.70 \rightarrow 4.68\times10^{36}\,\mathrm{erg\,s^{-1}}$), taking MoCHII from about 3% below the published 2.01–2.10×10^{37} at HII40 to slightly further below. The likely route is channel 1: $\alpha_1(\mathrm{H\,I})$ drops 0.52% from the Mao value to the Milne one, so the H I ground continuum recycles marginally fewer H-ionizing photons. This is not an argument against the change — the acceptance criterion is physical correctness, and an exact relation was being violated — but it is recorded rather than buried.

Two legacy options remain. `'exponential'` is the historical near-threshold approximation

$$E_{\mathrm{pkt}} = E_{\mathrm{th}} + kT_e[-\ln(\max(u,\epsilon))], \tag{29}$$

i.e. Eq. (27) with the $E^2\sigma_{\mathrm{pi}}$ weight dropped; it is kept to reproduce earlier runs. `'threshold'` emits at E_{th} and is a sensitivity bracket, not a model. The three thresholds are 13.598, 24.587, and 54.416 eV, and every model is a sampled photon energy, not a bin-center approximation.

Sampling weight. MoCHII emits equal-energy packets and fixes their number from the channel's *energy* luminosity. The packet energies must therefore follow the energy-luminosity density $E\,n(E)$, not $n(E)$ itself. With that pairing the represented photon rate is exactly the physical one,

$$N_{\mathrm{pkt}}L_{\mathrm{pkt}}\left\langle E^{-1}\right\rangle_{En} = R\left\langle E\right\rangle_n \frac{1}{\left\langle E\right\rangle_n} = R, \tag{30}$$

whereas drawing from $n(E)$ while counting packets from the energy luminosity inflates it by $\left\langle E\right\rangle_n\left\langle E^{-1}\right\rangle_n > 1$ (1.002 at 8000 K, 1.010 at 2×10^4 K). The CDF is tabulated in $x = (E-E_{\mathrm{th}})/kT$, interpolated in $\log T$ and inverted once, rather than inverting two neighboring temperature rows and interpolating the resulting energies.

Effect. Against the exponential approximation the converged HII40 calculation moves $r(\mathrm{He}^0 = 0.5)$ outward by 0.00063 pc and $r(\mathrm{C}^{2+} = 0.5)$ by 0.00052 pc, closing 0.3% of the residual against Cloudy c25. The two continua move in opposite directions because each follows its own mean energy: the H I continuum softens (14.460 → 14.427 eV at 10^4 K) since the hydrogenic cross section falls as E^{-3}, while the He I continuum hardens (25.449 → 25.460 eV) because the Verner He I fit falls more slowly near threshold. The exact spectrum is therefore a correctness fix, not the explanation of the Cloudy front residual. The same holds for the three He I cascade corrections described below: the two-photon shape moves $r(\mathrm{He}^0 = 0.5)$ by −0.00030 pc, the branch fractions by +0.00024 pc, and the fate of the 584 Å photon by −0.00363 pc. All four corrections to the emitted diffuse spectrum together move it −0.00306 pc, i.e. 1.7% of the 0.18 pc residual and *away* from Cloudy rather than toward it. Switching the whole fourth channel off moves it −0.103 pc, so the channel's presence matters and its internal detail does not.

Where the Cloudy front residual is not. Each of those four corrections is independently correct — three of them reproduce an external reference implementation and the first an exact detailed-balance relation — and the acceptance criterion is physical correctness, not agreement with a particular reference calculation, so a net displacement in the unhelpful direction is not an argument against any of them. What the sum does settle is the diagnosis: *the* 0.18 *pc residual is not in what the diffuse field emits.* Four independent corrections, spanning the emitted spectrum's shape, its branch weights and the fate of its strongest line, together account for 1.7% of the gap and with the wrong sign.

Bounding the remaining candidates. The quantity to be explained is $r(\mathrm{He}^0 = 0.5) = 4.12781$ pc against Cloudy c25.00's 4.30531 pc, i.e. −0.17750 pc after the α_1 correction above. The sensitivity runs below were measured against the pre-correction production point, 4.11970 pc; the 0.008 pc between the two baselines changes no lever, and the raw front positions are quoted as measured. The front's sensitivity to the diffuse field is calibrated two ways from the 584 Å measurement above: differentially (channel 4 lost 8.87% of its energy, is 17.6% of the diffuse field, and moved the front 0.00363 pc) and by switching that channel off entirely (17.6% of the field, 0.103 pc), giving 0.233 and 0.585 pc per unit fractional change in the total diffuse band luminosity. Two further sensitivity runs (HII40, 68 ranks, same Sobol stream, 38 iterations) scale the He I atomic data by 3%: reducing $\sigma_{\mathrm{pi}}(\mathrm{He\,I})$ moves the front +0.00799 pc and reducing the He I recombination rate moves it +0.02410 pc. Every candidate then has a number attached to it, and every number is too small:

axis	bound measured	to reach 0.178 pc
what the diffuse field emits	four corrections, net −0.00306 pc	wrong sign
explicit transport vs. on-the-spot	4.11970 to 4.18159 pc	the whole range is 0.062 pc
total diffuse band luminosity	0.233–0.585 pc per unit fraction	30–76%
$\sigma_{\mathrm{pi}}(\mathrm{He\,I})$	+0.00799 pc per −3% (0.27 pc/unit)	∼70%
$\alpha(\mathrm{He\,I})$	+0.02410 pc per −3% (0.80 pc/unit)	∼23%
α_A, α_1, $\alpha_B(\mathrm{He\,I})$	0.3%, 0.34%, 1.9% from Cloudy (§3.4)	already there; code spread is 11%
metal opacity	0.103 pc/unit upward, 0.615 downward (§4.7)	+180%, and saturating
metal recombination continua	0.29% of channel 2	∼0.001 pc
He I excited-level ionization	0.2–0.6% extra, and below the band floor	structurally absent
dust	`dust_model='none'` in the benchmark	excluded by construction
source photon budget	Q ratios 1.00005 and 1.00003	14% in $Q(\mathrm{He}^0)$
Cloudy-side resolution	median zone width 0.0106 pc at the front	17× finer already

The transport entry is the decisive one, because it was the leading candidate: switching the whole diffuse treatment between its extremes — explicit transport at 4.11970 pc and the on-the-spot approximation at 4.18159 pc — spans only 0.062 pc, and Cloudy sits 0.124 pc beyond the more favorable end of that range. No choice about how diffuse photons are transported can reach it. The photon budget is verified against the same 40-kK blackbody Cloudy was given: $Q(>13.598\ \mathrm{eV}) = 4.2656\times10^{49}\ \mathrm{s}^{-1}$ against 4.2653×10^{49} and $Q(>24.587\ \mathrm{eV}) = 4.6082\times10^{48}$ against 4.6081×10^{48}, ratios 1.00005 and 1.00003. Cloudy resolves the front with 40 zones between 4.0 and 4.6 pc, median width 0.0106 pc and maximum 0.039 pc, so the reference position is not a resolution artifact either.

The He-ionizing photon budget. Where the sensitivity lives is visible in the budget itself. Integrated over the He^+ zone of the converged HII40 model:

HII40	s^{-1}
stellar $Q(>24.587\ \mathrm{eV})$	4.6082×10^{48}
diffuse He I ground continuum, recycled	1.6135×10^{48}
total He-ionizing supply	6.2217×10^{48}
He^0 photoionizations = case-A recombinations	4.7008×10^{48}
absorbed by H^0 or metals, or escaped	1.5210×10^{48} = 24.4% of the supply

Nearly a quarter of the He-ionizing supply is lost to absorbers other than He^0, and halving that loss fraction alone would move the front 0.211 pc. The $\mathrm{H}^0/\mathrm{He}^0$ competition is therefore the one place where a few-percent difference has the leverage to matter, which is consistent with the radial signature below and with the metal-opacity mechanism of §4.7. The budget also closes geometrically: with the leaf volume computed correctly the summed gas volume matches the geometric shell 0.972–4.729 pc to 0.25%. (`LeafSize` in the output is the leaf *full* width, and the residual analysis script had used $(2\,\mathrm{size})^3$, eight times the cell volume. Because it enters only as a relative weight within a shell, every front position reported from it is unaffected and re-running after the fix reproduces all of them identically; the expression was nevertheless wrong and is now size^3. The table generator, the Python readers and the g1/g4 gates all used the convention correctly and were never affected.)

The radial signature. What the residual does have is a shape. The hydrogen structure agrees with Cloudy throughout the ionized zone; only the helium structure drifts, and it drifts monotonically outward:

r [pc]	1.6	2.0	2.5	3.0	3.5	3.8
$x(\mathrm{He}^0)$, MoCHII/Cloudy	1.007	1.015	1.031	1.059	1.109	1.196
$x(\mathrm{H}^0)$, MoCHII/Cloudy	0.981	0.984	0.989	1.000	1.005	1.009

A normalization difference in a rate or a cross section would be radius-independent; this one compounds with depth. That is what a 0.7% seed at 1.6 pc must do when He^0 carries almost all of the opacity above 24.6 eV: at 30 eV, $\kappa(\mathrm{He}^0)/[\kappa(\mathrm{He}^0)+\kappa(\mathrm{H}^0)]$ rises from 0.843 at 1.6 pc to 0.956 at 4.0 pc, so any excess He^0 shortens the mean free path of the very photons that would have removed it. The signature therefore identifies a small helium-specific seed, amplified geometrically, rather than a large error anywhere.

Closing 0.178 pc is 4.1% in radius and 13% in volume, which asks for $Q(\mathrm{He}^0)$ higher by $\sim$14% or $\alpha_B(\mathrm{He\,I})$ lower by $\sim$20%. $Q(\mathrm{He}^0)$ is verified to 0.003%, and MoCHII's α_A, α_1 and $\alpha_B(\mathrm{He\,I})$ now sit within 0.3%, 0.34% and 1.9% of Cloudy's own level-resolved values, with no published determination anywhere near 20% away. *The residual is therefore* *unexplained*, and what has been achieved is the systematic closing of the space of simple explanations: it is not in what the diffuse field emits, not in how those photons are transported, not in the He I photoionization cross section, not in the He I recombination rate, not in the metal opacity, not in the metal recombination continua, not in He I excited-level ionization, not in dust, not in the source photon budget, and not in the reference calculation's resolution.

The two candidates named in earlier revisions have both been examined and neither survives as a single parameter. The helium ionization *network* has been compared code to code (§3.4): every code sums the full cascade, MoCHII's $\alpha_A(\mathrm{He\,I})$ agrees with Cloudy's explicit level sum to 0.3%, and the He I excited levels that a resolved model atom carries are worth 0.2–0.6% of extra ionization. The metal opacity has been measured (§4.7): its upward derivative is 0.103 pc per unit, so closing the gap would need +180% against cross sections that reproduce Verner's to 10^{-6} and an Opacity Project comparison that runs 2–14% the other way.

What remains is the one hypothesis that no single-parameter experiment can reproduce: several differences of a few percent each — in the He I recombination rate, in the shape of $\sigma_{\mathrm{pi}}(\mathrm{He\,I})$, in the metal opacity of the 21–40 eV window, in Cloudy-side modeling choices not yet identified — acting together through the He^0 optical depth, which compounds them with depth exactly as the radial signature shows. Each is individually below its own measured lever; their product over 24.4% of the He-ionizing supply lost to competing absorbers is not bounded by any of the experiments above. Testing it requires varying several ingredients jointly, not one at a time, and that is the open item.

All diffuse packets carry the same luminosity as direct packets, $L_{\mathrm{pkt}} = L_{\mathrm{band}}/N_{\mathrm{pkt}}$, so their number is $N_{\mathrm{dif}} = \mathrm{nint}(L_{\mathrm{dif}}/L_{\mathrm{pkt}})$. A packet first selects a leaf from the CDF of $\sum_{\mathrm{ch}} L_{\mathrm{ch},l}$, then a channel from that leaf's four-channel luminosity weights. Its emission point is uniform in the selected cubic leaf and its direction is isotropic. It is then transported with the same exact-energy edge walk and path estimator as a stellar packet. Hence diffuse luminosity is represented by packet count, not by a different packet weight. At HII40 the diffuse band luminosity converges to 6.8×10^{38} erg/s $= 0.50\,L_\star$.

For output bookkeeping, the energy from Eq. (29) is mapped by binary search of `ion_eedge` (`ion_bin_of`). This works for both threshold-aligned and logarithmic diagnostic grids; an energy on an edge falls in the bin above it. In continuous mode this assignment does not modify E_{pkt} or its cross sections. In grouped mode only, the representative energy of that selected bin is used for transport. The binary search also corrects the former closed-form log-grid mapping, which could place a 24.6-eV He I recombination photon below the He I threshold on an aligned grid.

He I excited-level recombination (`par%hei_diffuse`). A fourth channel adds the He I *excited-level* recombination radiation (previously those photons simply vanished). Every case-B He II recombination reaches 1^1S through exactly one $n = 2$ term, so what the channel needs is not three absolute rates but three *branch fractions*; the absolute rate is already fixed by the $\alpha_B(\mathrm{He\,II})$ that sets the packet count. MoCHII takes all three from one internally consistent set of effective recombination coefficients,

$$
\begin{aligned}
\alpha_{\mathrm{eff}}(2^3\mathrm{S}) &= 2.10\times10^{-13}\,(T_4)^{-0.381},\\
\alpha_{\mathrm{eff}}(2^1\mathrm{S}) &= 2.06\times10^{-14}\,(T_4)^{-0.451},\\
\alpha_{\mathrm{eff}}(2^1\mathrm{P}) &= 4.17\times10^{-14}\,(T_4)^{-0.695},
\end{aligned}
\tag{31}
$$

in $\mathrm{cm}^3\,\mathrm{s}^{-1}$ — Benjamin, Skillman & Smits (1999) Table 1 with J. S. Mathis, in the form carried by K. Wood's `ionize` — and normalizes by their sum. That all three come from one source matters more than any individual value: a fraction assembled from fits with unrelated temperature dependences is not a fraction of anything. Normalizing by the sum is the source's own choice, made for the reason it states, that the fitting formulae carry different temperature dependences. Outside 3×10^3–2×10^4 K the fits are clamped, not extrapolated.

The closure

$$\alpha_{\rm eff}(2^3{\rm S})+\alpha_{\rm eff}(2^1{\rm S})+\alpha_{\rm eff}(2^1{\rm P}) = \alpha_B({\rm He\,II}) \tag{32}$$

is physically true, but MoCHII does *not* use it to derive a branch. Its two sides are independent atomic data: the three fits of Eq. (31) account for $0.90\,\alpha_B$ at 8000 K and $0.97\,\alpha_B$ at 10^4 K, and the ratio is reported at start-up rather than absorbed. Forcing it onto α_B would dump the entire mismatch between two data sets into whichever branch was solved for last. The resulting fractions, as the code prints them, are

T [K]	2^3S	2^1S	2^1P	$\sum\alpha_{\rm eff}/\alpha_B$
5×10^3	0.74084	0.07629	0.18288	0.767
8×10^3	0.76183	0.07591	0.16226	0.896
10^4	0.77121	0.07565	0.15314	0.970
1.5×10^4	0.78729	0.07507	0.13764	1.134
statistical	0.75000	0.08333	0.16667	—

against the AGN3 low-density statistical weights of the last row, which the code used before. The correction is a moderate temperature-dependent shift, not a reordering.

Two provenance points belong with these numbers. First, $2.06\times10^{-14}+4.17\times10^{-14} = 6.23\times10^{-14}$ exactly, so the $6.23\times10^{-14}(T_4)^{-0.827}$ that `nebcont_mod` took from MOCASSIN for the He I two-photon continuum (Section 4.17) is the *singlet total*, not $\alpha_{\rm eff}(2^1{\rm S})$. An earlier version of this section read it as the latter and closed 2^1P against α_B, obtaining 0.746 : 0.224 : 0.031 together with an argument that case-B trapping inverts the singlet split; both the numbers and the argument are withdrawn. Second, the 2^3S coefficient here is deliberately *not* shared with the `ALPHA3` of `hei_metastable_mod`, which is $2.10\times10^{-13}(T_4)^{-0.778}$ (Oklopcic & Hirata 2018): the same prefactor with a steeper exponent, 9% higher at 8000 K. That one is an absolute production rate for the metastable population; this one is a member of a set whose ratios are the branch fractions, and mixing sources would break the internal consistency those ratios depend on.

The 2^1S photons follow the tabulated Drake, Victor & Dalgarno (1969) distribution (`data/HeI2phot.dat`, the same table the nebular continuum uses) rather than a flat draw. That shape fixes the H-ionizing yield at 0.55644 photons and 8.9646 eV per decay, with a mean in-band energy of 16.110 eV, against the 0.56 and 17.109 eV of the flat surrogate. The two corrections are individually small and act in opposite directions. Per case-B He II recombination the channel emits

	H-ionizing energy [eV]	photons
statistical weights, flat two-photon	19.200	0.9634
statistical weights, Drake shape	19.149	0.9633
Eq. (31), Drake shape, 8000 K	19.223	0.9663

i.e. −0.27% in energy from the two-photon shape, then +0.39% in energy and +0.34% in photons from the branch fractions, for a net +0.12% and +0.30% against the statistical weights. In transport (68 ranks, same Sobol stream, branch fractions against the Drake-only state) the He I front moves by +0.00024 pc at HII40 and +0.00006 pc at HII20, $r(\mathrm{C}^{2+} = 0.5)$ by +0.00015 and 0.00000 pc, and $L(\mathrm{H}\beta)$ by +0.0121% and +0.0027%; both benchmarks still pass `tests/continuous_energy/check_hii_production.py`. Like the Milne ground continua, this is a correctness fix and not the explanation of the 0.18 pc Cloudy residual.

The fate of the 584 Å photon. The 2^1P branch decays to the ground state on He I Lyα, 584 Å = 21.22 eV, which is both an H-ionizing photon and a resonance photon in a medium optically thick to it in He^0. It therefore scatters many times before anything irreversible happens, and two irreversible outcomes compete: photoionization of a hydrogen atom (depositing $21.22-13.60 = 7.62$ eV of heat), or collisional transfer $2^1\mathrm{P} \to 2^1\mathrm{S}$ in the last He atom that absorbed it, after which the decay is the two-photon continuum instead. Which one wins is set by the H^0 and He^0 opacities the trapped photon sees, so it is a local, density- and ionization-dependent branching and not a constant. MoCHII resolves it with the escape-probability closure of Wood, Mathis & Ercolano (2004), Section 3.3,

$$p_{584} = \left[1 + \frac{77\, x(\mathrm{He}^0)}{\sqrt{T}\, x(\mathrm{H}^0)}\right]^{-1}, \tag{33}$$

the probability that the 584 Å photon ends up ionizing hydrogen (`hei_584_ionizes_H` in `hei_cascade_mod`). Its two limits are the correct ones: $x(\mathrm{H}^0) \to 0$ gives $p_{584} \to 0$, nothing left to ionize and everything converted, while $x(\mathrm{He}^0) \to 0$ gives $p_{584} \to 1$, nothing left to convert. K. Wood's `ionize` and CMacIonize carry the same closure in two algebraically different arrangements, and the gate `tests/hei_cascade/check_hei_584.f90` reproduces both to 2.2×10^{-16} over a grid in $(T, x_{\mathrm{H}^0}, x_{\mathrm{He}^0})$. The four codes in the reference trees sit at three different points on the question:

code	cascade coefficients	fate of the 584 Å photon	transported?
Wood `ionize`	Eq. (31), sum-normalized	p_{584} ionizes H; $1-p_{584}$ converts	no, on the spot
CMacIonize	the same set and normalization	the same split	no, on the spot
MOCASSIN	singlet total $6.23\times10^{-14}(T_4)^{-0.827}$	all converted ($p_{584} = 0$)	emissivity only
MoCHII	Eq. (31), sum-normalized	Eq. (33) split	yes, full transport

CMacIonize is also where MoCHII's flat-surrogate constant 0.56 came from, as $2 \times 28\%$ of the two-photon distribution above the Lyman limit; Eq. (31) and the Drake table replaced both of those constants above. MOCASSIN does not resolve the competition at all: its comment states the assumption that all He I singlets end in 2^1S, and the 6.23×10^{-14} singlet total is that $p_{584} \to 0$ limit written as one coefficient. MoCHII is now the only one of the four that applies the branching and then *transports* what it produces: Wood and CMacIonize destroy the 584 Å photon where it was created, so the deposition of whatever the branch yields is assumed local, while MoCHII carries the branch product through the grid with the same edge walk as any other packet and lets the grid decide where it is absorbed.

Because p_{584} depends only on the emitting leaf's own T, $x(\mathrm{H}^0)$ and $x(\mathrm{He}^0)$, the split folds into the branch weights of `hei_branch_energies` at build time and needs no random variate of its own, so the nine-coordinate Sobol assignment below is unchanged:

$$\begin{aligned} e(2^3\mathrm{S}) &= f_{2^3\mathrm{S}} \times 19.82\ \mathrm{eV}, \\ e(2^1\mathrm{P}) &= f_{2^1\mathrm{P}}\, p_{584} \times 21.22\ \mathrm{eV}, \\ e(2^1\mathrm{S}) &= \left[f_{2^1\mathrm{S}} + f_{2^1\mathrm{P}}(1-p_{584})\right] \times 8.9646\ \mathrm{eV}. \end{aligned} \tag{34}$$

A converted decay radiates the two-photon pair, 20.62 eV, and not 21.22 eV; the difference of 0.60 eV leaves as $2^1\mathrm{P} \to 2^1\mathrm{S}$ at 2.06 μm, which is non-ionizing and therefore correctly absent from the band. `gamma_2q_HeI` uses the same split for the He I two-photon nebular continuum, $\alpha_{2q} = \alpha_{\rm eff}(2^1\mathrm{S}) + [1 - p_{584}]\,\alpha_{\rm eff}(2^1\mathrm{P})$ (Section 4.17), which closes an internal inconsistency: `nebcont_mod` had inherited MOCASSIN's singlet total, the $p_{584} \to 0$ limit, while `diffuse_mod` transported every 2^1P decay as a 21.22-eV packet, the $p_{584} \to 1$ limit, so the two modules disagreed about the fate of the same decay. The domain of validity is recorded at the code site: this is an escape-probability closure for a 584 Å line assumed optically thick, static and dust free. Dust competes for the trapped photon and is not in the closure — a small correction at the Lexington dust content, not a small one at higher dust-to-gas ratios.

Effect of the 584 Å split. Weighted by the 2^1P emission rate on the converged grids, $\langle p_{584} \rangle = 0.117$ at HII40 and 0.0044 at HII20: 88% and 99.6% of that branch converts to the two-photon continuum, and the channel's output per case-B He II recombination falls from 19.22 to 17.47 eV at HII40 (–9.1%) and from 19.24 to 17.12 eV at HII20 (–11.0%), with the photon count down 6.6% and 7.9%. In transport (68 ranks, same Sobol stream, against the branch-fraction state):

	HII40	HII20
$\Delta r(\mathrm{He}^0 = 0.5)$ [pc]	–0.00363	–0.00022
$\Delta r(\mathrm{C}^{2+} = 0.5)$ [pc]	–0.00332	–0.00001
$\Delta r(\mathrm{H}^0 = 0.1)$ [pc]	–0.01002	(density bounded)
$\Delta L(\mathrm{H}\beta)$	–0.1997%	–0.0704%
$\langle T_e \rangle$ ($x_{\mathrm{H}^0} < 0.95$) [K]	9080.2 $\to$ 9053.4	8523.0 $\to$ 8522.3
diffuse band luminosity	–1.56%	–0.22%

Both benchmarks still pass `tests/continuous_energy/check_hii_production.py`. This is the largest of the four corrections to the emitted diffuse spectrum — six times the Milne ground continua and twelve times the branch fractions — and it still moves the He I front only 2% of the way to the Cloudy residual, in the wrong direction.

Three predictions were recorded before these runs, and grading them is worth keeping. The sign was predicted correctly: losing channel-4 photons pulls the fronts inward. The magnitude, of order 0.01 pc, was correct. The third, that HII20 would move more than HII40, was *wrong* — HII40 moved sixteen times further. The prediction had been read off p_{584} alone, which measures only what fraction of the 2^1P branch converts; front displacement instead scales with that branch's *absolute* contribution to the ionizing budget. Inverting the measured luminosity drops against the change per recombination puts channel 4 at 17.6% of the HII40 diffuse field and 2.0% of the HII20 one: a 20 kK spectrum makes few photons above 24.6 eV, so He^+ is scarce and the channel fed by its recombinations is small. A larger converted fraction of a much smaller channel moves the front less. The lesson generalizes — a local branching probability predicts what happens per event, never how far a front moves; that second factor is the channel's share of the budget, set by the hardness of the source.

With `launch_sequence='sobol'`, diffuse packets use a separate Owen-scrambled Sobol stream so that their changing count does not disturb the direct-source sequence. Its fixed nine coordinates are: one for the leaf CDF; three for the position within that leaf; two for (μ, ϕ); one for the channel CDF; one for the ground-continuum energy draw (or the excited-He branch); and one for the two-photon energy, drawn from the tabulated Drake shape, when that branch is selected. Coordinates that are irrelevant to a selected channel are still reserved, preserving the global packet-indexed layout across iterations and MPI task counts.

4.9 Solution-driven I-front re-refinement

At iteration `refine_iter` the octree is rebuilt from the current solution: an old leaf is flagged as front when $\epsilon < x_{\rm HI} < 1-\epsilon$ or a face neighbor differs by more than `refine_dx`; the new tree carries level `refine_lmax` wherever a cell overlaps a front-flagged leaf and `refine_lbase` elsewhere. The gas state and `rhokap` map over by position (each new leaf samples the old leaf containing its center), and every dependent array is recreated (the band opacity, tallies, metal Γ blocks, diffuse tables, rate arrays — the last was the one initially missed, an allocate-once array that faulted when $n_{\rm leaf}$ grew). All ranks rebuild identically from the shared state, and the old grid's shared-memory windows are *recycled*: `destroy_shared_mem` (the collective counterpart of `create_shared_mem`) frees the 18 old windows right after the position sampling — the old tree's last reader — so repeated re-refinement neither leaks nor holds old and new grids in memory simultaneously.

4.10 Dust in the ionizing band

The three parts of the dust coupling, now complete:

- **Absorption competition.** With `par%ion_add_dust`, each band bin adds $\kappa_{\rm d}(b,{\rm leaf}) = \rho\kappa_{\rm leaf}\,s_{\rm abs}(b)$ where $s_{\rm abs}(b) = (1-a(E_b))C_{\rm ext}(E_b)/C_{\rm ext}(\lambda_{\rm ref})$ comes from a kext table covering the EUV (the WD01/D03 R_V=3.1 table, 1 Å–$10^4\,\mu$m: $C_{\rm abs}/{\rm H} = 1.83\times10^{-21}$ at 13.6 eV falling to $3.96\times10^{-22}\,{\rm cm}^2$ at 100 eV). A trap worth recording: the D03 table is *not monotonic* in wavelength (an out-of-order seam) — without sorting, the linear interpolation search silently returned $s_{\rm abs}(100\,{\rm eV}) = 0.903$ instead of 0.832 and shifted the dusty Strömgren radius by +1.6%.

- **Scattering.** With `par%ion_dust_scatter` the scattering coefficient joins the extinction ($s_{\rm sca}(b) = a\,C_{\rm ext}/C_{\rm ext,ref}$; in-band Henyey–Greenstein asymmetry $g(E_b) = 0.67$–0.99 from the table $\langle\cos\rangle$ column) and interactions are sampled: after the analytic zero-variance direct walk, a first interaction is forced along the ray with weight $w_1 = 1 - e^{-\tau_{\rm edge}}$, the packet keeps the albedo fraction $a_{\rm eff} = \kappa_{\rm sca,d}/\kappa_{\rm ext}$ at each interaction, scatters into an HG direction, and continues with unforced flights (whose segments tally into J_b) under Russian roulette ($w_{\rm min} = 10^{-4}$). Stellar and diffuse packets share the walk; gas ionization, gas heating, and grain heating all derive from the same tally, so the energy split is bookkept by construction.

- **Live ionization-dependent dust with a PAH split.** `dust_model='laursen09_live'` recomputes the dust density from the *computed* $x_{\rm HI}$ every iteration, $n_{\rm d} \propto (Z/Z_{\rm ref})\,n_{\rm H}\left[(1-f_{\rm PAH})(x_{\rm HI}+f_{\rm ion}x_{\rm HII})+f_{\rm PAH}(x_{\rm HI}+f_{\rm ion,PAH}\,x_{\rm HII})\right]$, the PAH share carrying its own ionized-gas survival ($f_{\rm ion,PAH} = 0$ default: PAHs destroyed in ionized gas) — the generalization including its PAH-specific option.

- **Grain heating and the equilibrium temperature.** `Heat_dust(leaf)` = $\sum_b [4\pi J\Delta\nu]_b\,\kappa_{\rm abs,d}$ [${\rm erg\,s^{-1}\,cm^{-3}}$] is consumed by `dust_temp_mod`: $f_{\rm emit}(T) = 4\pi\int s_{\rm abs}(\nu)B_\nu(T)\,d\nu$ is precomputed from the same kext table (log-T

grid, 3.2–3200 K) and T_d(leaf) solves Heat_dust = $(\rho\kappa/d_\mathrm{cm})f_\mathrm{emit}(T_\mathrm{d})$; outputs are the T_d leaf map and the grid-integrated modified-blackbody IR spectrum (1–1000 μm). FUV grain heating is transported by the `par%add_fuv` band extension (Section 2.1): gas cross sections vanish below their thresholds automatically, so no new transport code is involved, and the two-segment grid keeps the ionizing bins at full resolution. This is the single equilibrium *mixture* temperature; the size- and material-resolved treatment is the SEDust coupling below.

4.11 Stochastic dust emission (SEDust coupling)

With `par%dust_sed`, `sedust_mod` (adapted from the MoCafe `dustemis_mod`) computes the grid-integrated dust SED with size-resolved stochastic heating and PAH bands, using the in-tree SEDust library (built from source under `SEDust/sed` by `build_lib.sh`; astrodust, DL07, and Zubko grain models). The coupling follows the MoCafe convention that makes the normalization exact by construction:

- **Shape from the local field.** For each leaf, the band tally J_b is converted to J_λ on the SEDust wavelength grid (1129 points, 0.0912–4×10^4 μm for astrodust). The band reaches into the EUV, below the SEDust optics grid; that energy is deposited into the shortest SEDust bin, conserving the absorbed power (the hardness of the stochastic temperature spikes is slightly underestimated there — a documented approximation). SEDust then returns the emission spectrum of the grain population in this field: equilibrium large grains plus the full temperature-distribution solve for small grains and PAHs.

- **Normalization from the transport.** The SEDust spectrum is used only as a shape: each leaf's emission is normalized to its absorbed power Heat_dust $\cdot\, V$ from the Monte Carlo tally (radiative equilibrium). H II regions are optically thin to the re-emitted IR, so no Lucy re-iteration is needed and the grid-integrated L_λ is accumulated directly (an `MPI_ALLREDUCE` over leaves distributed round-robin). $\int L_\lambda\, d\lambda = \sum$ Heat_dust V holds by construction and is written into the `_dustsed.txt` header.

- **Cost.** The exact stochastic solve costs $\sim$0.1–0.2 s for each leaf with nonzero grain heating; it runs once, at output time, after convergence (the dusty Strömgren smoke test: $\sim$1.9 hours on 8 ranks for a 64^3 grid — the solve, not the transport, dominates).

- **Smoke result.** Dusty Strömgren at 4×10^4 K: $\int L_\lambda d\lambda = 4.296\times10^{38}$ erg/s = $\sum$ Heat_dust V to 4 digits; the SED shows the PAH bands at 3.3/6.2/7.7/8.6/11.3/12.7/17 μm over a warm 20–60 μm plateau with the FIR falloff — the canonical stochastically heated spectrum for a hot compact field.

- **Live PAH share** (`par%sed_pah_live`): `dust_emission`'s channel output (astrodust: 'AD' + 'PAH') lets each leaf's spectrum be assembled with the PAH channel weighted by the leaf's PAH survival ($x_\mathrm{HI} + f_\mathrm{ion,pah} x_\mathrm{HII}$), divided under `laursen09_live` by the dust survival already carried by `rhokap`; the Heat_dust $\cdot\, V$ normalization keeps energy closure (the PAH energy shifts to the large grains). Smoke: the PAH bands drop to 0.26 (7.7 μm) and 0.15 (11.3 μm) of the PAH-everywhere reference, the FIR rises by 1.06–1.36×, and the 8-μm map turns from centrally peaked into a *ring* at the ionization front — the classic Spitzer/JWST bubble morphology.

4.12 Peel-off imaging

With `par%ion_peel`, one extra transport pass runs on the converged state with observer peeling (the MoCafe peeling-off convention, adapted to the band): every emitted packet — stellar and diffuse — peels a direct contribution $L_{\rm pkt} w\, e^{-\tau}/(4\pi r^2)$ toward each observer, and, when dust scattering is sampled, every interaction peels a scattered contribution with the Henyey–Greenstein phase function of the packet's bin (after the albedo weight, before the new direction is sampled). The optical depth to the box edge along the observer direction is integrated through `kap_ion` at the packet's bin by a tally-free walk (tallying the virtual peel ray would double count), so the images carry exactly the transport's extinction: gas, dust absorption, and dust scattering. Images are TAN projections on the MoCafe observer geometry (`obsx/y/z` or angle lists, rotation matrix, auto-sized pixel scale), written as surface brightness [erg s^{-1} cm^{-2} sr^{-1}] split into direct/scattered and ionizing/FUV channels. The band tally is rebuilt by the imaging pass, so the written rates reflect the same field the images carry. With `par%peel_bins` the channel axis becomes the band bin — `direct_cube`/`scatt_cube` (n_x, n_y, n_ν) — for hardness maps and synthetic EUV/FUV photometry (the band-integrated channel images are still written, derived from the cube; the gate shows the expected color: the FUV scattered halo shrinks from a 15.5-px to a 12.0-px half-light radius between 6.2 and 13.2 eV).

External-field peel. The direct peel of an external packet (Section 2.2) replaces the isotropic $1/(4\pi r^2)$ with its Lambert emission density about the entry-surface inward normal $\hat{n}_{\rm in}$, $p(\Omega) = (\hat{k}\cdot\hat{n}_{\rm in})/\pi$, so the contribution is $L_{\rm pkt}\, w \max(\hat{k}_{\rm obs}\cdot\hat{n}_{\rm in}, 0)\, e^{-\tau}/(\pi r^2)$: only the far-side surface (inward normal toward the observer) contributes, and the image is the background field seen through the medium — the cloud in silhouette. In the optically thin limit the image-integrated flux is $F = JA_{\rm proj}/d^2$, and with `save_direc0` the ratio `direc0/direct` $= e^{+\tau}$ is again the line-of-sight optical-depth map. The scattered peel is origin-agnostic and unchanged.
For the *dust* bands no transport is needed — the nebula is optically thin to its own IR — so `par%dust_emis_bands` instead stores SEDust band emissivities of each leaf (`_dustemis`, the same block layout as the `_emis` file) and the Python map maker projects them into flux-conserving dust-band images exactly like line maps.

4.13 Cartesian (car) grids

`par%grid_type = 'car'` builds a single-level Cartesian grid and stores its leaves in raster order: leaf = cell = $1 + i_x + n_x(i_y + n_y i_z)$. No tree topology, no neighbor table, and no geometry arrays are built. (`'uniform'` is accepted as a backward-compatible alias.) The leaf list comes either from a file (`par%amr_file`, single level) or — with `amr_file` empty — directly from the namelist: `par%nx/ny/nz` cells over the box $[-x_{\rm max}, x_{\rm max}]\times\ldots$ (cubic cells required).

Uniform density and sphere/shell geometry. A gas-density model is applied on the leaf centers before either grid (`'car'` or `'amr'`) is built: `par%nH_const` overrides every leaf's $n_{\rm H}$ with one uniform value, and `par%rmax/rmin` zero $n_{\rm H}$ outside a sphere of radius $r_{\rm max}$ / inside $r_{\rm min}$ (a shell). Because the cut is on the leaf centers, a sharp sphere or shell edge can be resolved by the octree (`'amr'` with a refined leaf list) while the density stays exactly uniform — the case that motivates uniform density on `'amr'` as well as `'car'`.

Analytic geometry. All geometry queries go through accessor functions in `octree_mod` (`cell_cx/cy/cz/ch`, `leaf_cell`, `leaf_cx/cy/cz/half`). On the octree these are plain array reads; on the car grid they decode the raster index and evaluate $c_x = x_{\min} + (i_x + \frac{1}{2})\,\Delta$ — the same floating-point expression that previously filled the arrays, so nothing changes numerically. The car build therefore allocates *none* of the tree arrays: not only `children` ($8\times$), `neighbor` ($6\times$), and `parent`, but also the cell centers, half-widths, and the leaf$\leftrightarrow$cell maps. At 512^3 this removes the entire $\sim$10 GB of shared tree memory; the grid costs only its physical fields. All physics modules (thermal, ionization, diffuse, lines, dust) read geometry exclusively through the accessors, so they run unchanged on either grid.

Traversal (`par%car_walk`). The car grid is not a physics choice — both walks discretize the same transport integral and agree to rounding — so `par%car_walk` selects only *how* the ray is stepped:

- `'dda'` (default): dedicated incremental Amanatides–Woo walks, the natural car traversal. Each ray precomputes, once, the parametric distance to the next face crossing per axis ($t_{\max,x/y/z}$) and the crossing period ($t_{\Delta,x/y/z} = \Delta/|k_{x/y/z}|$); the inner loop advances by comparisons and additions alone — no exit-face solve at each step, no leaf re-lookup. The tally expressions (analytic first-flight $w(e^{-\tau_{\rm in}} - e^{-\tau_{\rm out}})/\kappa$, pathlength sums, τ-target stop) are identical to the `'amr'` walks. It is the fastest mode: $\times1.64$ over the `'amr'` walk, $\times2.0$ over the octree (total Strömgren run time $13.0 \to 6.5$ minutes).
- `'amr'` (verification): the octree band walks (`raytrace_ion_to_edge_amr`, `_to_tau_amr`, `_tau_only_amr`) run unchanged; only their two geometry primitives branch on the grid mode — the leaf finder `amr_find_leaf` becomes three integer divisions (`car_index`) and the face-neighbor step `amr_next_leaf` becomes pure integer arithmetic on the raster index (`car_next`). The exit face and distance are still recomputed each step from the cell center and half-width (`amr_cell_exit`), exactly as on the octree. Because the floating-point order along each ray is then unchanged, the same-seed Strömgren run is *bit-identical* to the single-level-octree run ($\max|\Delta x_{\rm HI}| = 0$) — the cross-check the DDA walk cannot provide (it differs at rounding by design, $\max|\Delta x_{\rm HI}| = 2.7\times10^{-15}$, $V_{\rm ion}$ identical). This certifies the raster backend rather than serving production (it is $\times1.25$ over the octree, slower than DDA).

Everything downstream is untouched (the gas modules only consume leaf-indexed arrays) and `refine_front` is octree-only.

4.14 Plane-parallel slab illumination

An *xy-periodic slab* models a medium that is horizontally infinite: a packet that leaves an x or y face re-enters at the opposite face, while the z faces escape. The medium may vary only in z (a 1D column, `nx = ny = 1`) or be an arbitrary 3D tile (`nx, ny` > 1) repeated periodically. The wrap is source-agnostic — it is a boundary condition on the transport, so an internal source under `xy_periodic` is a horizontally periodic array of sources, and the external slab illumination below rides on the same wrap. In the octree walks the wrap re-enters the packet at the opposite face's exact boundary coordinate (not by $x \pm x_{\rm range}$, which can land a rounding-epsilon outside the box and have the leaf lookup drop the packet); the incremental DDA wraps the integer index (`modulo`), the parametric t-coordinates being translation-invariant on a uniform grid.

External illumination (`source_geometry='slab'`) enters the top ($+z$) and/or bottom ($-z$) face, each as a collimated beam at incidence angle θ or an isotropic (Lambert) field. The physical normalization is the flux crossing the horizontal face,

$$F_z = \int_{\text{hemisphere}} I_{\text{in}} \cos\theta \, d\Omega = \begin{cases} \mu F_n, & \text{beam } (\mu = \cos\theta,\ F_n \text{ normal to the beam}) \\ \pi I, & \text{isotropic } (I \text{ the specific intensity}), \end{cases} \tag{35}$$

so the user inputs the source strength (F_n or I) and the geometric factor (μ or π) gives F_z. The entering luminosity of one periodic tile is $L = F_z A_{\text{zface}}$ (the packet weight is L/N_{phot}); A_{zface} cancels in every per-volume result, which therefore scales linearly with F_z and is independent of the tile area and lateral resolution. The two faces compose additively (both lit $\Rightarrow L = (F_{z,\text{top}} + F_{z,\text{bot}})A$).

The emergent observable is the boundary intensity: a packet escaping a z-face carries its surviving luminosity into a ($\mu = |k_z|$, face) histogram, written as $I(\mu)$ [erg s^{-1} cm^{-2} sr^{-1}] with $F = 2\pi \int I \mu \, d\mu$, together with the reflected/transmitted/absorbed energy budget. Both the direct beam (the analytic edge walk) and, with dust scattering, the scattered/diffuse flights contribute, so the budget closes to unity. Point-observer peel-off is disabled under `xy_periodic` (the periodic images admit no finite-distance observer). The quasi-random (`sobol`) launch covers the slab in both modes (a six-dimensional layout: frequency, face, the two in-face coordinates, and the Lambert μ, ϕ).

4.15 PDR-zone physics

Three switches (default off) extend the FUV zone beyond ion structure:

- `par%metal_ne`: electrons from the metal cascade join the n_e closure of the cell solve, $n_e = n_{\text{H}}[x_{\text{HII}} + y_{\text{He}}(\ldots)] + \sum_{\text{el}} A_{\text{el}} n_{\text{H}} \sum_i (i-1) f_i$ — a $\lesssim 10^{-3}$ correction inside the H II region but the *only* electrons beyond the front ($n_e \simeq n_{\text{CII}} \approx A_{\text{C}} n_{\text{H}}$ in the PDR zone), where they control the [C II]/Mg II excitation and the photoelectric efficiency.

- `par%metal_heat`: metal photoheating enters the thermal balance; the heating integrals $\mathcal{H}_i = \sum_b [4\pi J \Delta\nu]_b \sigma_i (1 - E_{\text{th},i}/E_b)$ are accumulated together with the metal Γ's.

- `par%grain_pe`: grain photoelectric heating and grain recombination cooling with the Bakes & Tielens (1994) fits, $\Gamma_{\text{pe}} = 1.3\times10^{-24} \epsilon G_0 n_{\text{H}}$ with $\epsilon = 4.87\times10^{-2}/[1 + 4\times10^{-3} x^{0.73}] + 3.65\times10^{-2}(T/10^4)^{0.7}/[1 + 2\times10^{-4} x]$, $x = G_0\sqrt{T}/n_e$, and $\Lambda_{\text{rec}} = 4.65\times10^{-30} T^{0.94} x^{\beta} n_e n_{\text{H}}$, $\beta = 0.74/T^{0.068}$, both scaled by `pe_scale` times the leaf's dust amount relative to Milky-Way dust, measured from the leaf opacity itself: $(\rho\kappa/d_{\text{cm}})/(n_{\text{H}} C_{\text{ext,MW}}(V))$ with $C_{\text{ext,MW}}(V) = 4.868\times10^{-22}\,\text{cm}^2/\text{H}$ — robust against the DGR-versus-`cext_dust` input conventions (the first gate run had DGR = 1 with the MW extinction folded into `cext_dust`; a naive DGR/0.01 scale made the heating 100$\times$ too strong and drove the zone to a false Lyα-capped 12 kK equilibrium). G_0 is the local FUV field in Habing units summed from the FUV bins of the tally (`add_fuv` required). Three further guards, each found by the gate: (i) the BT94 fits are evaluated at $\min(T, 10^4\,\text{K})$ — outside their fit range the $T^{0.7}$ term of ϵ diverges; (ii) both grain terms carry an x_{HI} weight — the BT94 charging parameter knows only the FUV field, so in ionized gas (EUV-charged grains, grain energy already routed to the IR via `Heat_dust`) it would overheat badly; (iii) the package includes the *H-impact* fine-structure cooling of [C II] 158 μm and [O I] 63 μm (two-level, low-density limit; $q_{ul}^{\text{H}} = 7.6\times10^{-10}(T/100)^{0.14}$ and $9.2\times10^{-11}(T/100)^{0.67}\,\text{cm}^3\,\text{s}^{-1}$, Barinovs et al. 2005) — the cooling fits are electron-impact only, and with $n_{\rm HI}/n_e \sim 10^3$–10^4 in the PDR zone the H collisions ARE the coolant. Without (iii) the zone has no coolant between the fine-structure and Lyα regimes, and the $\epsilon(x)$–n_e feedback (collisional ionization raises n_e, neutralizing the grains and raising ϵ fifty-fold) drives it to a false $\sim 10^4$ K equilibrium.

Scope note: this is the atomic PDR skin — no H_2/CO chemistry, no line transfer, no dust–gas collisional coupling — so PDR temperatures are first-order estimates, adequate for the ion structure and rough [C II] emissivities.

4.16 n-level solver and line luminosities

`nlevel_mod` evaluates exactly the recipe of Section 3.10 (Clenshaw evaluation of the Chebyshev series) and solves the statistical equilibrium $\dot{n} = 0$, $\sum_i n_i = 1$ (partial-pivot elimination; de-excitation $q_{ul} = 8.629\times10^{-6}\,\Upsilon/(g_u\sqrt{T})$, excitation by detailed balance). `lines_mod` integrates $L_{\rm line} = \sum_{\rm leaf} j\, n_{\rm ion} V$ on the converged state and normalizes to the SH95 Hβ. The Fortran chain was verified against the Python reference solver on an identical state: 26 line luminosities agree to four significant digits. With `par%emis_output` the same solve also writes the emissivities leaf by leaf (`_emis`: an `emis_<el>_<stage>` block for each ion and `emis_h` for the SH95 lines, [erg s^{-1} cm^{-3}] with $\sum \mathrm{emis}\cdot V = L_{\rm line}$, verified to 4×10^{-4}%), together with `n_H`, n_e, T_e, the H/He fractions, and the metal stage fractions — a self-contained input for line maps and position–velocity work.

4.17 The two-photon continuum: emission, cooling, transport

The metastable $2s$ (2^1S in He I) levels decay by simultaneous emission of two photons sharing the transition energy. That one channel enters the code in three separate places, and the three treatments are deliberately *not* the same; they are collected here because the differences are easy to mistake for inconsistencies.

Emission (`nebcont_mod`, Section 3.11). The emission coefficient is written per $n_e n_{\rm ion}$ and is fed by the *recombination* cascade, one term for each of the three ions:

ion	weight	ν_0 [Ryd]	$\alpha_{\rm eff}(2s)$
H I	$n_e n_{\rm HII}$	0.7496	$0.8368\times10^{-13}(T_4)^{-0.723}$ (Pengelly 1964)
He I	$n_e n_{\rm HeII}$	1.514	$\alpha_{\rm eff}(2^1{\rm S}) + [1-p_{584}]\,\alpha_{\rm eff}(2^1{\rm P})$ (BSS99)
He II	$n_e n_{\rm HeIII}$	3.00	SH95 table, $6.16 \to 2.04\times10^{-13}$ over 5–30 kK

with the Nussbaumer & Schmutz (1984) shape $A(y)$, $y = \nu/\nu_0$, for H I and (scaled by $Z^6 = 64$ in both A and A_{2q}) for He II, and the Almog & Netzer (1989) tabulated shape for He I. The He I entry has a history worth recording. The port took MOCASSIN's $6.23\times10^{-14}(T_4)^{-0.827}$, which is the recombination coefficient to *all* singlets — the exact sum $2.06\times10^{-14} + 4.17\times10^{-14}$ of the 2^1S and 2^1P coefficients of Eq. (31), refitted as a single power law, so the exponent -0.827 is a fit to the sum and not a sum of exponents. Using it is MOCASSIN's stated assumption that every He I singlet ends in 2^1S, i.e. the $p_{584} \to 0$ limit of Eq. (33), and it disagreed with the $p_{584} \to 1$ limit `diffuse_mod` was transporting. That constant is no longer used. `gamma_2q_HeI` now builds the emitting 2^1S population from the two coefficients separately,

$$\alpha_{2q}(\mathrm{He\,I}) = 2.06\times10^{-14}(T_4)^{-0.451} + [1-p_{584}]\,4.17\times10^{-14}(T_4)^{-0.695}, \tag{36}$$

with the same local p_{584} that `diffuse_mod` folds into its branch weights, so the written continuum and the transported field now describe one nebula (Section 4.8). The MOCASSIN constant is recovered exactly as $p_{584} \to 0$ and $\alpha_{\rm eff}(2^1{\rm S})$ alone as $p_{584} \to 1$. Only H I carries the collisional suppression

$$\gamma_{2q} \to \frac{\gamma_{2q}}{1 + n_e q_{2s\text{-}2p}/A_{2q}}, \qquad A_{2q} = 8.2249\ {\rm s}^{-1}, \tag{37}$$

with $q_{2s\text{-}2p} = 5.31\times10^{-4}$ (at $T \leq 10^4$ K) falling to $4.71\times10^{-4}\,{\rm cm}^3\,{\rm s}^{-1}$ (at $T \geq 2\times10^4$ K), so the continuum is quenched above $n_{\rm crit} \simeq 1.6\times10^4\,{\rm cm}^{-3}$. No factor is applied to the helium terms: A_{2q} is $51.3\,{\rm s}^{-1}$ for He I 2^1S and $526\,{\rm s}^{-1}$ for He II $2s$, putting their critical densities one to two orders of magnitude higher. The three terms are summed into the total L_ν of `_nebcont.txt` and also written separately as its last column. Because the population is the recombination cascade alone, collisional excitation of $2s$ contributes nothing to the emitted continuum, which is therefore a lower bound in the partly neutral gas of an ionization front.

Cooling (`cooling_mod`, Section 3.9). The H I line-cooling fit is the $n_e \to 0$ statistical-equilibrium limit of the 25-level CHIANTI atom, and level 2 of that atom *is* $2s$ ($82258.956\,{\rm cm}^{-1}$, $A = 8.229\,{\rm s}^{-1}$, the two-photon rate) with the $1s \to 2s$ collision strength present. Collisional excitation of $2s$ therefore cools with its full 10.2 eV, and the leading term of the fit, $A_1 = 4.4262\times10^{-17}$ at $T_1 = 1.1844\times10^5$ K, is exactly that excitation temperature.

The cooling must *not* carry the density quenching the continuum does, and does not: proton and electron l-mixing converts $2s$ to $2p$ and the two-photon continuum into Lyα, but the gas loses the same 10.2 eV on either route. (CHIANTI `h_1` tabulates no $2s \leftrightarrow 2p$ collision strength, so the equilibrium solve leaves $2s$ unquenched — harmless for the same reason. Nor does the level produce a spurious line: `lines_mod` extracts only the Hα/Hβ/Hγ windows of the collisional component, so nothing is double counted against the continuum.) Helium has no collisional-excitation line cooling at all — `cooling_total` carries recombination, free-free, collisional ionization and the H I line term only — so collisional two-photon cooling of He I 2^1S and He II $2s$ is absent, as it is in MOCASSIN, TORUS, Wood et al., and CMACIONIZE.

Transport (`diffuse_mod`, Section 4.8). Only helium two-photon photons can ionize anything: the H I continuum stops at 10.2 eV, so omitting it costs nothing. The He I 2^1S branch is one of the four diffuse channels and carries its H-ionizing fraction 0.55644 from the Drake table (Section 4.8; the flat surrogate's 0.56 is no longer used). The He II continuum, which extends to 40.8 eV and can ionize both H and He I, is *not* transported, and neither is He II Lyα — of no consequence wherever He^{2+} is absent (both Lexington benchmarks), an omission to close for the hard planetary-nebula and AGN spectra of Section 6..

5. Validation gates and results

5.1 Rate integrals vs. analytic attenuation

Uniform sphere ($n_{\rm H} = 1$, $R = 1$ pc, $x_{\rm HI} = 0.1$, $y_{\rm He} = 0.1$), central 10^5 K source, 16 bins, 10^8 packets. Against the cell-volume-averaged analytic attenuation with identical bins: $\Gamma_{\rm HI}$ mean bias +0.004%, median 0.38% (pure Monte Carlo noise), max 2.6%; $\Gamma_{\rm HeI}$ bias +0.011% (Figure 1).

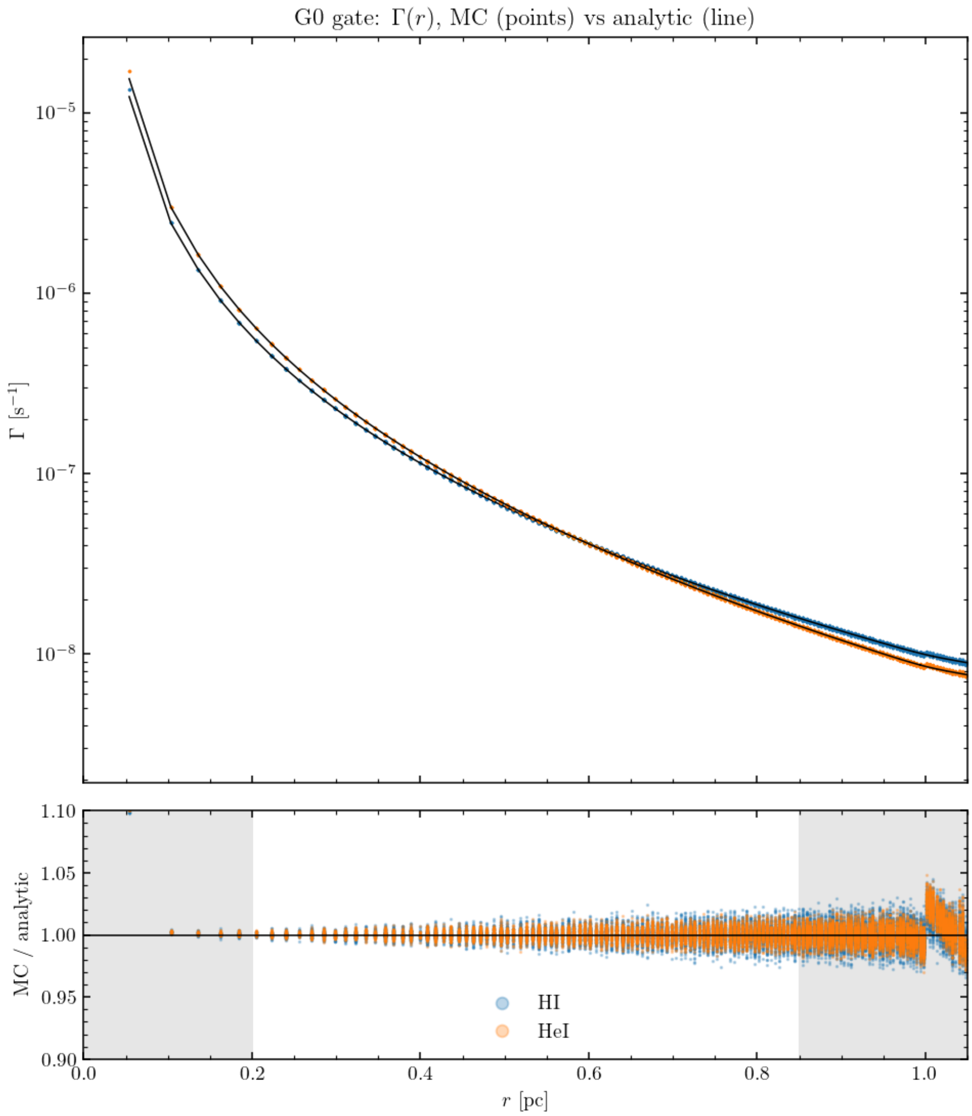

Figure 1: Rate-integral gate: $\Gamma(r)$ from the Monte Carlo tally (points) against the analytic point-source attenuation (lines); the lower panel shows the ratio (shaded regions excluded: cell-size effects at the center, the voxelized sphere edge outside).

5.2 Strömgren sphere

Uniform $n_{\rm H} = 100$ sphere, $Q_{\rm H} = 10^{49}\,{\rm s}^{-1}$ ($T_\star = 4\times10^4$ K, 32 bins), case B, $T_e = 10^4$ K. The reference is an exact 1D radial marching with identical bins and rates — necessarily *Gauss–Seidel* (each shell solved with the already-updated interior opacity): a Jacobi sweep advances the front only one absorption length per pass and never converges. Sharp-edge analytic $R_S = 3.153$ pc; the 1D reference gives $R_{\rm eff} = (3V_{\rm ion}/4\pi)^{1/3} = 3.0365$ pc (the $\sim$4% gap is physical: He consumes photons and the model re-emits none).

run	leaves	$R_{\rm eff}$ [pc]	vs. 1D
uniform level 6	262 144	3.0534	+0.56%
front-refined levels 4–6	74 320	3.0534	+0.56%
center-refined levels 4–6	7 568	3.1078	+2.35%

The front-refined octree reproduces the uniform grid to four decimals with 3.5× fewer cells; the center-refined control (coarsest cells *at* the front) shows why the I-front is the one place needing resolution (Figure 2).

5.3 Thermal H/He nebula

Same configuration with the thermal balance on (Figure 3): $T_e(r)$ matches the 1D thermal reference to median 0.136%, mean bias +0.006%, max 0.95% over $0.3 < r < 2.8$ pc — including the central He III-zone temperature structure and the front drop. Without metals the nebula runs at 19–24 kK, the known pure-H/He overshoot that motivates the metal frame.

5.4 The MOCASSIN Lexington benchmarks

HII20/HII40: blackbody 20/40 kK, $Q_{\rm H} = 1.006/4.26\times10^{49}\,{\rm s}^{-1}$, $n_{\rm H} = 100$, hollow shell $R_{\rm in} = 3\times10^{18}$ cm, abundances He 0.1, C 2.2×10^{-4}, N 4×10^{-5}, O 3.3×10^{-4}, Ne 5×10^{-5}, S 9×10^{-6}. The stored MOCASSIN 2.02.73.2 baselines are *reduced* runs (10^5 photons, 3 iterations) — at iteration 3 only 27% of cells have converged. To close the comparison MOCASSIN itself was re-run converged (2×10^6 photons, 40 iterations, 72% cells; a finer nx=25 grid confirms resolution), and MoCHII with the *corrected* charge exchange of Section 3.8.

Temperatures. T_e weighted by $n_{\rm HII}n_e$, MoCHII with its current defaults (threshold-aligned grid, corrected charge exchange of Section 3.8, and the density-suppressed cooling of Section 3.9); the benchmark inputs carry the ionizing-band luminosity that reproduces the specified $Q_{\rm H}$, transport the diffuse field, and use case-B recombination lines:

	MoCHII (`local_ne`)	MoCHII (`low_density`)	CLOUDY c25.00	Lexington-2000 (Ercolano+03)	MOCASSIN 40-it
HII20	6925 K	6376 K	6910 K	6402–6749 K	—
HII40	8169 K	7894 K	8210 K	7720–8199 K	8176 K

With the density-suppressed cooling default (`par%cooling_model = local_ne`) MoCHII gives $T_e = 8169$ K (HII40) and 6925 K (HII20), matching a current CLOUDY c25.00 run (8210 and 6910 K) to 0.50% and 0.22%. Both modern codes sit a few hundred K *above* the 2000-era published ranges of Ercolano et al. (2003) — $\langle T[N_pN_e]\rangle = 7720$–8199 K

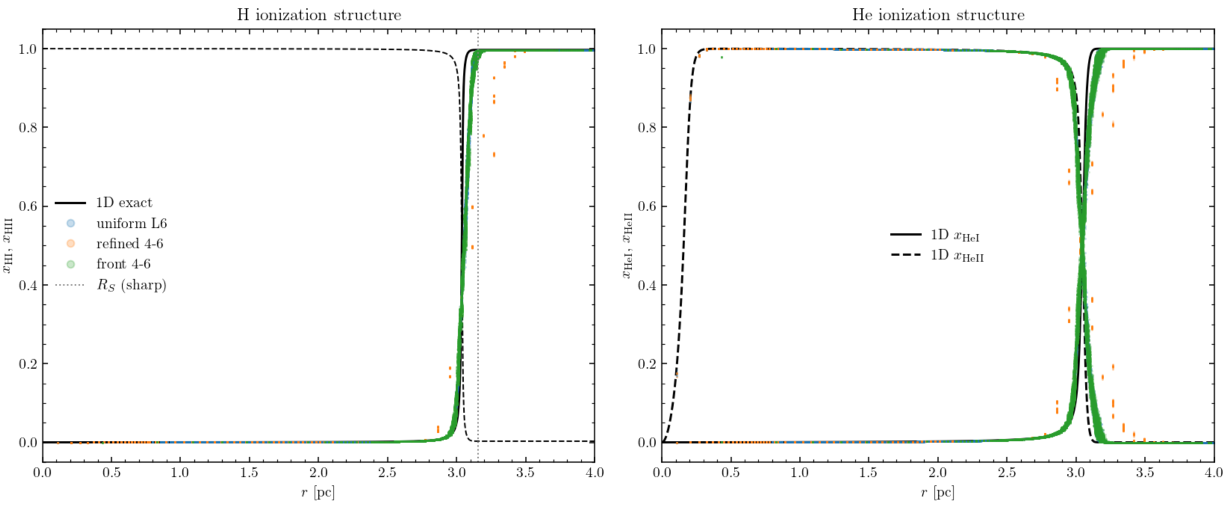

Figure 2: Strömgren gate: H and He ionization structure against the 1D exact reference (black); uniform level-6 (blue), center-refined (orange), front-refined (green, overlying blue).

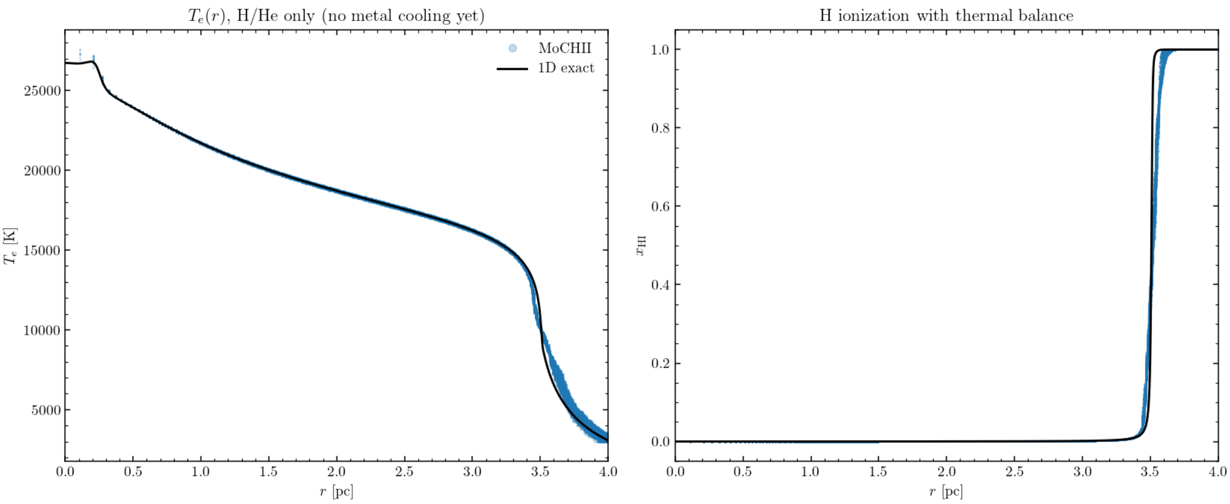

Figure 3: Thermal gate: $T_e(r)$ and $x_{\rm HI}(r)$ with the thermal balance, against the 1D exact reference.

(median 8026) at HII40 and 6402–6749 K (median 6662) at HII20 — whose tabulated columns carry older atomic data; the two modern codes now agree with each other far more closely than either does with the historical spread.

The $\sim$9% HII40 excess reported in earlier revisions (a superseded T_e = 8913 K) is *resolved*: it was two defects in the Tier-1 cooling fits (Section 3.9), compounded by the then-missing density suppression. **(D1)** The ground-term fine-structure populations were distributed by Boltzmann factors — the *high*-density limit — instead of the $n_e \to 0$ limit, which suppressed the far-infrared fine-structure cooling ([O III] 52/88 μm, [N III] 57 μm, [C II] 158 μm) by factors 1.6–2.9. **(D2)** The `.scups` Burgess–Tully *scaling* energy was used as the physical transition energy in both the Boltzmann factor and the radiated photon, in place of the observed `.elvlc` level difference (e.g. [O II] cooling low by 0.58× at 8 kK). Both were located by a fixed-state cooling-budget comparison against CLOUDY c25.00 (its `save cooling each` output): D1 was 23.9% and D2 10.6% of the HII40 cooling budget, while missing channels ([S IV] 10.5 μm at 1.2%, now added; all others < 0.2%) and trace-species coupling (n_e plus metal photoheating, together +0.2%) were excluded. Regenerating the fits from the $n_e \to 0$ statistical-equilibrium limit with observed energies gives HII40 7894 K and HII20 6376 K (the `low_density` option); restoring the density suppression at the local n_e (the `local_ne` default) raises them to 8211 and 7025 K, and the subsequent corrections to the diffuse field — the Milne ground continua, the He I cascade and the α_1 correction of §4.8 — settle them at 8169 and 6925 K. The former “cooling-side residual” was thus the low-density over-cooling of the low-$n_{\rm crit}$ fine-structure lines, and there is no open temperature discrepancy with the current reference code.

Ionization structure (n_e-weighted means, HII40). The He and metal ionization is set correctly once the ionizing grid resolves the He I edge. With `par%ion_align_edges` on (the default) or nnu_ion $\geq$ 128 the He-ionizing photon budget returns to the exact blackbody value ($Q_{\rm HeI}/Q_{\rm HI}$ = 0.108 against 0.127 for a 32-bin log grid), He I rises from 0.058 to 0.163 (MOCASSIN 0.207, closing most of the gap), and the metal 3+ fractions fall correspondingly. A separate in-code swap of MOCASSIN-style recombination (Pequignot 1991 RR + Nussbaumer–Storey 1983 DR) showed the recombination-coefficient vintage to be a minor lever: MoCHII’s Badnell 2023 α(X III $\to$ X II) runs $\sim$0.8× MOCASSIN’s, but the swap moved the fractions by only $\sim$0.026 and closed $\sim$17% of the ionization gap. The table below is the old 32-bin log grid, the reference point for the alignment discussion (corrected CX vs. the converged 40-iteration MOCASSIN):

ion	MoCHII (log)	MOCASSIN	ion	MoCHII (log)	MOCASSIN
C II	0.188	0.355	O II	0.344	0.491
C III	0.805	0.640	O III	0.653	0.496
N II	0.180	0.386	Ne II	0.524	0.587
N III	0.818	0.600	Ne III	0.474	0.401
S II	0.194	0.066	S III	0.750	0.786

Switching on the aligned-grid default brings the He and metal fractions substantially closer to MOCASSIN — He I 0.058 $\to$ 0.163, O III 0.653 $\to$ 0.601, C III 0.805 $\to$ 0.694 — so most of the apparent over-ionization in the table is the log-grid discretization at the He I edge, not a physics error. At HII20 the CX correction is negligible (low excitation): the structure matches everywhere except Ne I/II (the 21.6-eV threshold at a 20-kK source, a known code-spread hotspot) and the S II/III split.

Line fluxes (ratios to Hβ). With the corrected cooling the optical collisional lines — formerly systematically high because T_e ran $\sim$9% hot — now fall inside the published spread: [O III] 5007 drops from $\sim$4.3 to $\sim$2 against the published 1.64–3.27, and the [N II], [S II] and [Ne III] ratios move into range. Absolute $L(\mathrm{H}\beta) \approx 1.9 \times 10^{37}$ (HII40) and $4.7 \times 10^{36}\,\mathrm{erg\,s^{-1}}$ (HII20) sit $\sim$3% below the published $2.01\text{–}2.10 \times 10^{37}$ and $4.83\text{–}5.04 \times 10^{36}$, consistent with helium absorption of the >24.6-eV photons and the weak temperature dependence of α_{B}; with the corrected cooling the HII40 front is captured within the prescribed sphere (outermost-shell $x_{\mathrm{HII}} = 0.11$, recombinations 95% of Q_{H}). The emissivity chain itself is exact (Section 4.16).

5.5 Re-refinement gate and the TNG demonstration

The re-refinement of Section 4.9 on the Strömgren configuration (level-5 start, rebuild to level 7 at iteration 15, 32 768 $\to$ 464 248 leaves):

run	leaves	R_{eff} [pc]	vs. 1D
re-refined 5$\to$7	464 248	3.0432	+0.22%
native uniform level 7	2 097 152	3.0430	+0.21%

The two agree to +0.008% with 4.5× fewer leaves, and both improve on level 6 (+0.56%) — resolution-convergent, as the resolution-convergence requirement expects. The front profiles overlie exactly.

The post-processing demonstration runs MoCHII on an Illustris-TNG 60-kpc cutout (the MoCafe converter output, n_{H} and x_{HI} columns from the simulation) with a $Q_{\mathrm{H}} = 10^{53}\,\mathrm{s^{-1}}$, 40-kK source at the density peak, thermal balance + metals + re-refinement (Figure 4): dense clumps cast sharp neutral shadows, the circumgalactic gas sits at $x_{\mathrm{HI}} \sim 10^{-4}$ with n_e-weighted mean $T_e = 7502$ K, and the refinement (32 768 $\to$ 617 135 leaves) tracks the *distributed* ionization topology of a turbulent medium — clump skins and partially ionized gas, not a single thin shell. The run also demonstrates why the max-cell convergence test stalls in such media (clump-surface jitter); a volume-integrated criterion is queued.

5.6 Dusty Strömgren gate

With the MW D03 dust of Section 4.10 added to the Strömgren configuration ($\tau_{\mathrm{d,abs}}(13.6\,\mathrm{eV}) = 1.7$ over the dust-free radius), against the 1D reference carrying the identical dust absorption in every bin:

case	R_{eff}(1D) [pc]	R_{eff}(MC) [pc]	dev
dust-free	3.0365	3.0534	+0.56%
full MW dust	2.1983	2.2058	+0.34%
`laursen09_live`	3.0218	3.0425	+0.68%

The classic dusty reduction ($R/R_0 = 0.724$ in 1D, 0.722 between the two MC runs) is reproduced within the gate tolerance; the live ionization-dependent dust drives itself to near dust-free ($R/R_0 = 0.996$: interior dust vanishes with the computed x_{HII}), confirming the dust–ionization feedback loop converges self-consistently. Energy budget: dust absorbs 61% of the band luminosity in the full-dust case ($L_{\mathrm{abs}} = 1.933 \times 10^{38}$ of 3.178×10^{38} $\mathrm{erg\,s^{-1}}$).

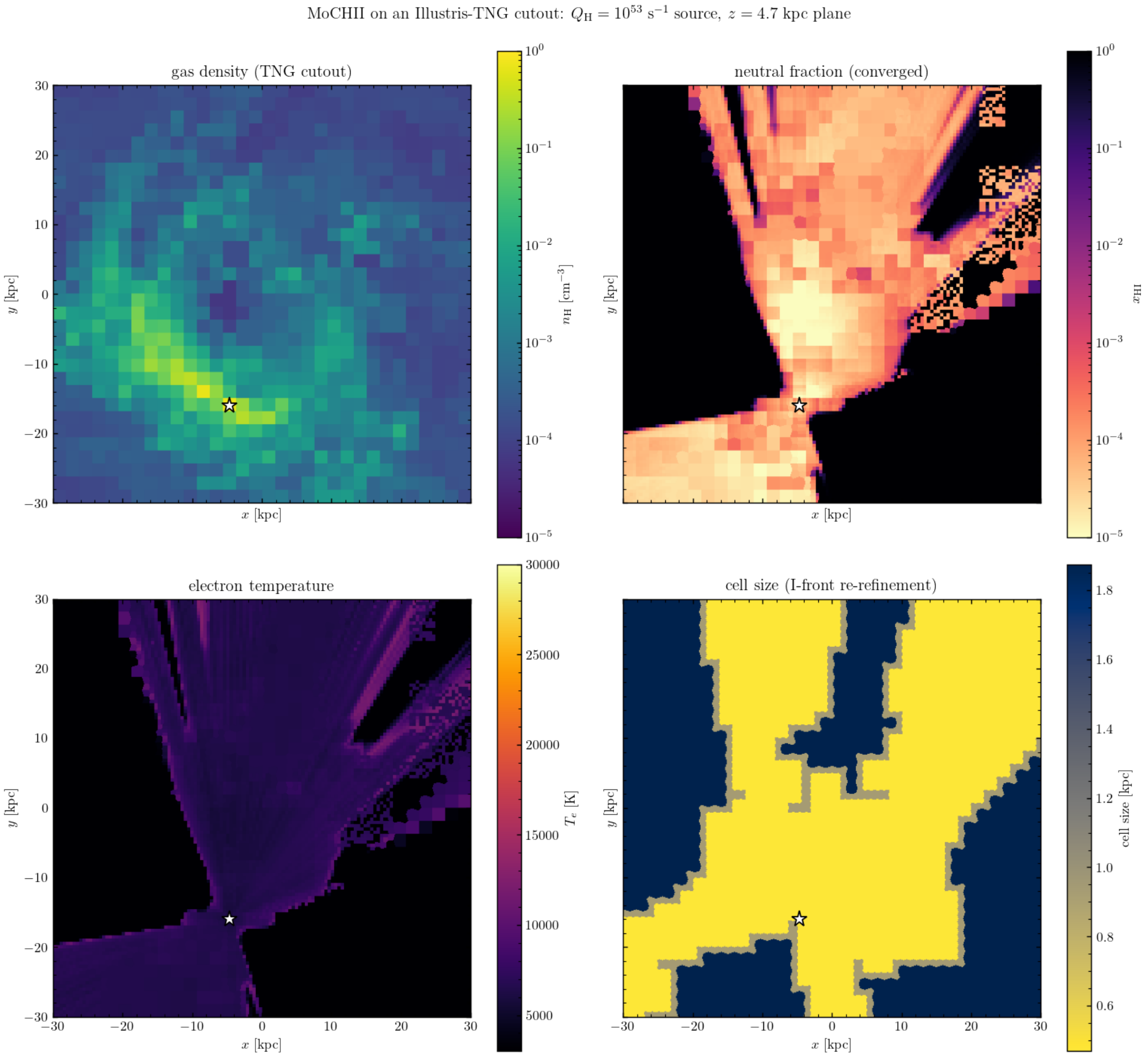

Figure 4: MoCHII on an Illustris-TNG cutout: gas density, converged neutral fraction (note the clump shadows), electron temperature, and the re-refined cell size, in the source plane.

Scattering bracket. Turning the scattering on must land the sphere between two analytically defined bounds: treating the dust extinction entirely as absorption (scattered photons destroyed; 1D $R_{\rm eff} = 2.059$ pc) and the absorption-only treatment (scattered photons never interact; 2.198 pc, MC counterpart 2.206 pc). The full-scattering run gives $R_{\rm eff} = 2.197$ pc — inside the bracket and close to the absorption-only bound, exactly as the strong forward scattering ($g = 0.67$–0.99) predicts: forward-scattered photons continue nearly radially, and only the small diffused fraction is lost from the radial budget. The dust-absorbed luminosity rises by 0.7% (rescattered photons reabsorbed) — energy bookkeeping consistent.

Dust temperature and IR. With the band widened to 6 eV (the band luminosity rescaled by the exact blackbody ratio 2.0969 so $Q_{\rm H}$ is preserved) and scattering on, the equilibrium mixture temperature runs from $T_{\rm d} = 87$ K at 0.3 pc to 28 K at the sphere edge — the canonical HII-region gradient — and the IR spectrum integral closes on the absorbed power to $L_{\rm IR}/L_{\rm abs} = 1.000$. (This demo predates the two-segment `add_fuv` grid and rescaled the single-segment band by hand; the +1.5% bin-coarsening shift it measured is what the two-segment grid now removes.)

5.7 FUV extension gate: PDR-side C II and Mg II

The `add_fuv` smoke test (dusty Strömgren, 32 ionizing + 16 FUV bins, thermal + metals with Mg/Fe, dust scattering on; `tests/d_dusty/d_fuv_opt.in`) verifies the band mechanics and the physics it was built for. Mechanics: the two-segment grid reports $L_{\rm FUV}/L_{\rm ion} = 1.0969$ for the 4×10^4 K Planck source at a 6 eV edge — exactly the blackbody band ratio applied by hand in the earlier demo — with $Q_{\rm H}$ preserved by construction. Physics: inside the front (2.2 pc with this dust and cooling) the metals sit at C III/Mg III as before; *beyond* it, where H is neutral and the ionizing bins are exhausted, the FUV bins propagate under dust extinction alone and hold the low-threshold stages fully singly ionized — radially averaged C II = 1.000 and Mg II = 1.000 from the front out to the sphere edge, the PDR-side structure that the ionizing-only band cannot produce (there the same zone recombines to C I/Mg I). Two documented limitations in that zone: electrons from metal ionization are not fed back into n_e (the trace-species design), and with photoheating ≈ 0 the thermal bisection pins T_e at `te_min` — PDR temperatures require photoelectric heating, outside the present scope. The front-zone cells with vanishing heating also dominate the max $|\Delta T_e|$ convergence test (another argument for the queued volume-integrated criterion); the H II-interior structure and T_e (8.1–10.5 kK) converge normally.

5.8 External field and multiple sources

The source-component model of Section 2.2 is gated in the `ext_field`, `multi_src`, and `peel_ext` directories under `tests/`:

- **External field.** In an optically thin box ($\tau \sim 7\times10^{-3}$) the interior band-integrated mean intensity recovers $J_{\rm ext}$ to 0.9981 (`'rec'`) and 0.9991 (`'sph'`), uniform and isotropic (axis asymmetry $< 0.1\%$); the entering power equals $\pi J A_{\rm surf}$ [Eq. (2)] exactly.
- **Superposition.** Two point sources against the analytic sum $\sum_i L_i/(16\pi^2 r_i^2)$: median deviation 0.76%, mean ratio 0.9989; a mixed point + external run, median 0.75%. Two-temperature additivity — a two-source run against the sum of the two single-source runs — agrees to a bin-wise median of 0.12% and to 1.0001 in the band total.
- **Spectra.** A multi-column `src_spectrum_file` tabulating the same Planck curves as its `src_tstar` twin repro-

duces it to 0.010% in the worst bin; `ext_spectrum` against its `ext_tstar` twin to 0.019% (this check also caught and fixed the single-component path ignoring `ext_tstar/ext_spectrum`).

- **External peel.** The thin-box image-integrated direct flux against $F = JA_{\mathrm{proj}}/d^2$: –0.22% (`'rec'`); the `direc0` flux +0.15% at chord $\tau \sim 2$; the measured central-pixel $\tau_b = -\ln(\mathrm{direct}/\mathrm{direc0})$ reproduces $n_{\mathrm{H}}\,\sigma_{\mathrm{HI}}(E_b)\,\ell$ bin by bin to four digits with the exact hydrogenic cross section (deviating from the E^{-3} power law by up to $\sim$10%, as σ_{exact} does).
- **Physical-unit spectra.** A `'per_ev'` source against its `'shape'` twin agrees to 0.004% in the worst bin (the derived `src_lum` is exact); the `'per_ang'` and `'per_um'` wavelength conversions reproduce it to 0.013%; rescaling a physical file to a set scale matches its shape twin to 0.0000%. A physical `ext_spectrum` J_E file recovers the interior $\langle J\rangle/J = 0.998$, and doubling `ext_intensity` doubles the field exactly (2.0000).
- **ISRF presets.** The measured FUV energy density matches the analytic value to $u_{\mathrm{FUV,meas}}/u_{\mathrm{FUV}} = 1.0002$–$1.0003$: 8.94 (Draine), 5.23 (Habing), 6.01×10^{-14} erg cm^{-3} (Mathis), with exactly zero in the ionizing bins.
- **Backward compatibility.** Single-source runs are bit-compatible with the pre-change binary (worst difference 1 ULP, 9.9×10^{-16} relative).

5.9 Plane-parallel slab

The slab of Section 4.14 is gated under `tests/slab/` against the analytic plane-parallel transport, with the `xy_periodic` wrap exercised throughout:

- **xy-periodic wrap.** A tall thin periodic column ionizes fully ($\langle x_{\mathrm{HII}}\rangle = 0.9999$) where the isolated (non-periodic) control reaches only 0.7072 — the wrap re-injects the lateral flux; the octree (`car_walk = 'amr'` / `grid_type = 'amr'`) and the DDA agree to 8.5×10^{-17} for `nx`$\geq$2, and after the exact-face wrap fix (Section 4.14) also at `nx = 1` (1.3×10^{-18}). The AMR leaf-list and car density-cube file inputs both run correctly under `xy_periodic`.
- **Beam attenuation.** A frozen neutral slab illuminated by a collimated beam gives a photoionization-rate profile $\Gamma_{\mathrm{HI}}(z)$ that is a *perfect* single exponential ($R^2 = 1.00000$); at $\theta = 60^\circ$ ($\mu = 0.5$) the decay slope is exactly slope_0/μ (ratio 1.0000) — the $e^{-\tau/\mu}$ law and the $\cos\theta$ convention are correct. The isotropic mode gives $F_z = \pi I$ exactly and a shallower monotone profile.
- **Emergent $I(\mu)$ and energy budget.** A normal beam transmits $e^{-\tau_\perp} = 0.8874$ exactly (budget closes, 0.88739 + 0.11261 = 1.0), with $I(\mu)$ a single spike at $\mu = 1$ and zero reflection (no scattering); a $\theta = 60^\circ$ beam transmits $e^{-\tau/0.5} = 0.7874$ exactly; isotropic illumination gives a broad $I(\mu)$ over all bins (mean $\mu = 0.583$, forward-shifted by the $e^{-\tau/\mu}$ weighting). Two-face illumination is a symmetric superposition.
- **Photoionization slab.** A full ionization solve (case B OTS) builds a plane-parallel ionization front at the Strömgren depth $d = \Phi/(n_{\mathrm{H}}^2\alpha_B)$; the front position matches to within $\sim$5–7% (recombination-coefficient and finite-front-width uncertainty) and scales with the incident flux ($d \propto F_z$, ratio 1.94 for a factor-of-two change, the small deficit being the roughly constant front width).

5.10 Argon as a data operation

Adding argon touched no algorithm: two cooling fits (Ar III, Ar IV), two n-level files, one element file (VFKY96 rows Z=18, Voronov, Badnell RR + SVS82 DR2, KF96 CX), one abundance parameter. PyNeb cross-check: [Ar III] 7136/7751 within 0.7%; [Ar IV] 4740/4711 within 4.0% (a density pair — the same database-spread class as [O II] and [S II]; PyNeb uses RB97/RGJ19 there). End-to-end smoke test: [Ar III] 8.99 μm, 7136, 7751 appear in the line table with 7136/7751 = 4.17, the A-ratio, as expected.

5.11 Magnesium and iron

Mg and Fe repeat the recipe with three new data wrinkles, all absorbed by the readers: the Fe II power-law RR (`RR2`), the SVS82-form DR (`DR2`), and the Mg/Fe charge-exchange forms and stage attribution (Section 3.8). Products: cooling fits Mg II (0.05% max error), Fe II (0.11%), Fe III (0.42%); n-level files Mg II (3 levels), Fe II (13 levels, the IR/optical forbidden lines), Fe III (14 levels); element files with VFKY96 Z=12, 26 rows. PyNeb cross-check: the [Fe III] 4658/4702 ratio agrees to −6% to −16% — the largest gate deviation so far, and a known property of iron: PyNeb's default Fe III collision strengths (BB14) differ from CHIANTI v11's by tens of percent (the iron collision-strength spread in the literature), compounded by the 14-level truncation of a 34-level model. The number is recorded as a data-vintage bracket, not a code error — the same chain reproduces [O III] to 0.6%. For iron-line studies `par%fe_levels_full` loads expanded 16/34-level Fe II/III models (`nlevel_fe_{2,3}_full.txt`, 298 [Fe III] lines vs 38, e.g. 4658/4702 +3%) in place of the compact defaults. End-to-end smoke test (HII20 sphere with $n_{\rm Mg}/n_{\rm H} = 3\times10^{-6}$, $n_{\rm Fe}/n_{\rm H} = 3\times10^{-7}$, depleted gas-phase values): Mg loads 3 stages, Fe 4 stages; the line table gains the Mg II 2796/2803 doublet (ratio 1.98, the A-ratio limit) and the [Fe II] 1.257 μm, 5.34 μm, and 26 μm lines.

5.12 Silicon, chlorine, calcium (five stages)

Si, Cl, and Ca bring the registry to eleven elements, each tracked through five stages. The one framework extension was chlorine's cross section (the VY95 path of Section 3.1 — Cl is absent from the VFKY96 outer-shell table); Si carries the KF96 charge exchange from the MOCASSIN `chex` rows (recombination direction), Cl and Ca have no transcription available. n-level files (all $\leq$ 0.65% worst transition): Si II/III/IV, Cl II/III/IV, Ca II, Ca V; Cl V and Ca III/IV have no CHIANTI v11 level data and are tracked without lines (the Ar II pattern). Cooling fits: Si II 1.1%, Si IV 0.2%, Ca II 0.4%; Si III 12% and the Cl ions 8–9% resist the multi-exponential form — accepted, since at $\sim3\times10^{-7}$ abundances their share of the thermal balance is $\sim10^{-4}$ and the diagnostics run through the exact n-level solve. PyNeb gates: [Cl II] 8579/9124 to 1.5%; Si III] 1892/1883 to 6.9% and [Cl III] 5518/5538 to 6.6% (PyNeb's DK94/BZ89 compilations against CHIANTI v11 — the established database-spread class). End-to-end smoke: [Si II] 34.8 μm enters the line table at 4.4% of Hβ (the strongest metal line after [C II] — the ionization-front/PDR coolant it is), with Si III] 1883/1892, Si IV 1394/1403, [Cl II] 8581/9126, and the [Cl III] pair at 5518/5538 = 1.402, the low-density limit.

5.13 Line and continuum outputs

The converged-state outputs now include: the SH95 H I recombination lines (smoke-state check: Hα/Hβ = 2.94 at $\sim$7 kK); the n-level collisional lines for the 24 registry ions (including [C II] 158 μm, C III] 1909, [N III] 57 μm, Mg II

2796/2803, and the [Fe II] IR lines); and the grid-integrated nebular continuum L_ν (free-bound + free-free from the verified tables and the Milne extension above the edges, plus the three two-photon continua), written on a 400-point log-ν grid from 11.4 μm to 217 Å.

5.14 He I 2^3S metastable and 10833 Å absorption

The 2^3S diagnostic (Section 3.13) is gated in `tests/hei_meta/` on three runs (low/high density and an `add_fuv` run). **Stage 1**: the data-file analytic limits reproduce the low-density ratio $\alpha_3 n_e/A_{31}$ = 1.6509×10^{-7} (spec 1.65×10^{-7} at n_e = 100, 10^4 K), the plateau 6.513×10^{-6}, and $n_{\rm crit}$ = 3945 cm^{-3} (the n_e-scan half-plateau crossing lands at 3926); the Fortran module n_3 matches the independent Python closed form on each run's own state to worst relative 4×10^{-15} (machine precision); the low-density core sits within median 2.8% / max 3.9% of the radiative limit and the high-density run ($n_e = 1.1\times10^4 > n_{\rm crit}$) at 0.676 of the plateau; Φ_3 from the band tally matches an independent J_ν integration to 1.5×10^{-9} (max 9.07×10^{-5} s^{-1}); and $\alpha_3(10^4\,\mathrm{K}) = 0.998\times0.75\,\alpha_B(\mathrm{He\,II})$, the diffuse `HEI_F3S` consistency. **Stage 2 (10833 absorption)**: the LINE10833 constants carry the f ratio 1:3:5 and $A = 1.0216\times10^7$ s^{-1} each (NIST/p-winds, not the CHIANTI 1.15×10^7); a synthetic uniform cube reproduces the column conservation $\sum NA_{\rm pix} = \sum nV$ and the interior column $n_3\,\ell$ exactly, and the line-center $\tau_0 = \int\kappa_0\,dl$ per component and for the doublet matches the hand integral $\kappa_0^{\rm hand}\,\ell$ to $< 2\times10^{-16}$; the optically-thin case raises no warning while a $\times10^5$ scaled case pushes $\tau_0 > 1$ and fires it. On the high-density run the column conserves to 9×10^{-15} and the doublet/J'=0 τ_0 ratio is 8.003, equal to the $(f_1+f_2)/f_0$ = 8.002 oscillator-strength ratio of the blended lines (that run's metastable column is optically thick, $\tau_0\sim10^4$ — flagged by the warning, as designed).

6. Known limitations and next steps

- By default He recombination emits only the three ground continua; the He I excited-channel ionizing photons ($\sim$60% of He recombinations) are re-emitted only when `par%hei_diffuse` is set — their effect on the He I/II split against MOCASSIN is small (the split is set by the $>$24.6-eV budget). The He II excited cascade has no such channel: its Lyα (40.8 eV) and two-photon continuum can ionize H and He I but are not transported (Section 4.17). Immaterial wherever He^{2+} is absent, as in both Lexington benchmarks; it belongs to the hard-spectrum track.
- The EUV part of the leaf field, below the SEDust optics grid, is folded into the shortest SEDust bin (energy-conserving; spike hardness slightly underestimated).
- The thermal-loop line cooling is suppressed to the local n_e by default (`cooling_model = local_ne`) through the (T, n_e) table of Section 3.9, and the H-impact [C II] 158 μm / [O I] 63 μm term (`metal_cooling_H`, weighted by $x_{\rm HI}$) carries its own two-level saturation. What remains is the *cold* limit: the cooling fits hold over 10^3–10^5 K and collapse below that floor (their lowest excitation temperature is 2054 K for C I, far above the 23.6 K of [C I] 609.1 μm), and no H-collider term exists for the [C I] fine-structure lines — so gas colder than $\sim10^3$ K, where hydrogen is the collider, is outside the cooling model. Ar II carries no cooling or lines (no CHIANTI v11 level data).

- PDR zones: with the Section 4.15 switches off (the default), the FUV extension gives the PDR-side *ion structure* only; with them on, n_e, metal photoheating, and photoelectric heating make the PDR temperatures first-order estimates — H_2/CO chemistry and line-transfer cooling remain out of scope.
- The cell-max convergence test stalls at the front-cell Monte Carlo noise floor (and at clump-surface jitter in turbulent media); the volume-integrated criterion (`conv_crit = 'vol'`) resolves this and is the recommended production setting.
- The MOCASSIN Lexington benchmark temperatures are resolved (Section 5.4): with the density-suppressed cooling default MoCHII (HII40 8169 K, HII20 6925 K) matches a current CLOUDY c25.00 run to 0.50% and 0.22%, the earlier excess having been traced to two Tier-1 cooling defects (D1/D2) and the missing density suppression, all fixed. The one remaining note is that both modern codes sit a few hundred K above the 2000-era published Lexington temperatures — a data-vintage difference, not a code defect.
- The HII40 He I front sits 0.178 pc inside the CLOUDY c25.00 one and that difference is unexplained (Section 4.8). Every single-parameter candidate is now excluded by measurement, including the two that were still open — the helium ionization network (Section 3.4) and the metal opacity (Section 4.7). The untested hypothesis is a combination of several few-percent differences compounding through the He^0 optical depth, which needs a joint rather than a one-at-a-time experiment.
- The registry-wide recombination audit (Section 3.7) found no defect — no transcription error, and no form a better source could replace — but it leaves one question open: the rates that form *neutral* atoms differ between Badnell and Verner & Ferland (1996) by 10–16%, and for the metals no third determination is available to say which is right. Adjudicating it needs an independent determination for each ion (R-matrix C I/O I/Ne I recombination, or CHIANTI totals not derived from Badnell). For He I, where such a determination does exist, MoCHII is within 0.3% of it.
- Further metal ions as science needs them.